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V810i S2EX Panel Handling Subsystem





Panel Handling Subsystem

Automated Board Inspection

Hardware and software that:

1. Load the panel into the system
2. Detect the panel presence inside the system
3. Unload the panel

Board Flow Direction: Left to Right

1. Close inner barrier
2. XY stage move to the loading position
3. Adjustable Rail open to the correct width (Panel Width)
4. Panel Clamp will open (Bottom clasper move down)
5. Extend PC, Board Stop Right
6. Left outer barrier will open
7. Left Panel Clear Sensor will detect the board during loading
8. Conveyor belt will turn on
9. The panel will slow down once reach the 2nd Optical PIP Sensor (Slow Sensor)
10. The panel will stop once reached the 4th Mechanical PIP Sensor (Stop Sensor) and conveyor belt turn off.
11. Retract the PC, Board Stop Right
12. Panel Clamp will close (Bottom Clasper move up)
13. Left Outer barrier will close
14. The stage will move to check with the Filter Height Sensor (Applicable for High Magnification)
15. Open inner barrier

Board Flow Direction: Left to Right

1. Close inner barrier
2. XY stage move to the loading position
3. Adjustable Rail open to the correct width (Panel Width)
4. Panel Clamp will open (Bottom clamper move down)
5. Extend PC, Board Stop Right
6. Waiting for the signal from the SMEMA
7. Left Outer barrier will open
8. Left Panel Clear Sensor will detect the board during loading
9. Conveyor belt will turn on
10. The panel will slow down once reach the 2nd Optical PIP Sensor (Slow Sensor)
11. The panel will stop once reached the 4th Mechanical PIP (Stop Sensor) and conveyor belt turn off.
12. Retract the PC, Board Stop Right
13. Panel Clamp will close (Bottom Clamper move up)
14. Outer barrier will close
15. The stage will move to check with the Filter Height Sensor (Applicable for High Magnification)
16. Open inner barrier

Panel Inspection Sequence (>16.4 inch)

Board Flow Direction: Left to Right

After finish inspecting 1st region of the board, reposition for the 2nd region inspection

1. Move the stage to the right outer barrier
2. Move down the bottom clamber to un-clamp the panel
3. Turn on the conveyor belts (Left to Right)
4. Detect the panel at the right panel clear sensor
5. Turn off the conveyor belts
6. Extend the PC, Board Stop Left
7. Reverse the conveyor belts direction (Right to Left)
8. The panel will slow down once reached the 3rd Optical PIP Sensor (Slow Sensor)
9. The panel will stop once reach the 1st Mechanical PIP Sensor (Stop Sensor) and conveyor belt turn off
10. Retract the PC, Board Stop Left.
11. Move up the bottom clamber to clamp the panel.
12. Continue 2nd region inspection



Automated Board Inspection

Panel Unload Sequence

1. Close the inner barrier
2. XY Stage move to the unloading position
3. Move the clamber down to un-clamp the panel
4. Turn on the conveyor belt
5. Move the panel to the unloading side Panel Clear Sensor
6. Panel Clear Sensor detect the Panel
7. Turn off the conveyor belt
8. Outer barrier opens
9. Waiting for the panel to be removed from the system
10. Panel Clear Sensor become clear
11. Outer barrier close

1. Close the inner barrier
2. XY Stage move to the unloading position
3. Move the clamber down to un-clamp the panel
4. Turn on the conveyor belt
5. Move the panel to the unloading side Panel Clear Sensor
6. Panel Clear Sensor detect the Panel
7. Turn off the conveyor belt
8. Outer barrier opens
9. SMEMA will detect the board
10. Panel Clear Sensor status become clear after unload the board.
11. Outer Barrier Close.

ABI GUI Interface - Digital I/O tab

Automated Board Inspection

Input from Sensor
as actual position
feedback

Command provided
to mechanism
hardware

v810 Service - Subsystem Status and Control - 5.11.1 (10706) (Built 07/26/17 at 11:18:05)

File Edit Service Tools Actions Window Help

Display Measurements in: Inches (KV: 0.0 uA: 0.0)

Confirmation, Diagnostics, & Adjustments Subsystem Status & Control X-ray Safety Test

Main Panel Handling Panel Positioner X-ray Source X-ray Camera Optical Camera Ethernet-Based Light Engine Light Tower SMEMA Interlocks Motion Control Digital IO Connected Processors Hardware Simulation Control

Show Hardware Configuration Description Initialize

Input Bits (Read Only)

☒ Left Outer Barrier Closed	☒ Right Outer Barrier Closed	☐ Left Outer Barrier Open	☐ Right Outer Barrier Open
☐ Left Panel Clear Sensor Blocked	☐ Right Panel Clear Sensor Blocked	☐ Right PIP Optical Stop Sensor Engaged	☐ Left PIP Optical Stop Sensor Engaged
☐ Optical Stop Controller Installed	☐ Interlock Service Mode Off	☒ Interlock Powered On	☒ Interlock Switch ID Bit 5
☒ Interlock Switch ID Bit 4	☐ Interlock Switch ID Bit 3	☒ Interlock Switch ID Bit 2	☒ Interlock Switch ID Bit 1
☐ Interlock Switch ID Bit 0	☒ Inner Barrier Closed	☐ Inner Barrier Open	☒ Z-Axis At Ready Safe Position On
☒ Z-Axis Servo On	☒ X-Ray Left Safety Level Sensor On	☒ X-Ray Right Safety Level Sensor On	☐ Downstream System Ready To Accept Panel
☐ Panel From Upstream System Available	☐ Optical Stop Controller Done	☐ Optical Stop Controller Error	☐ Optical Stop Controller Enabled
☐ Left Panel In Place Sensor Extended	☒ Left Panel In Place Sensor Retracted	☐ Right Panel In Place Sensor Extended	☒ Right Panel In Place Sensor Retracted
☐ Optical PIP Sensor Reset On	☐ Panel Clamps Opened	☐ Z-Axis At Low Mag Position 1 On	☐ Z-Axis At Low Mag Position 2 On
☐ Z-Axis At High Mag Position 1 On	☒ Motorized Height Level Sensor - Low Mag	☐ Motorized Height Level Sensor - High Mag	☒ Motorized Height Level Sensor - No Alarm
☒ X-ray Cameras Connected	☒ X-ray Cameras Powered On	☒ X-ray Cameras Power In Range	☒ X-ray Cameras Phase Locked

Output Bits (Write)

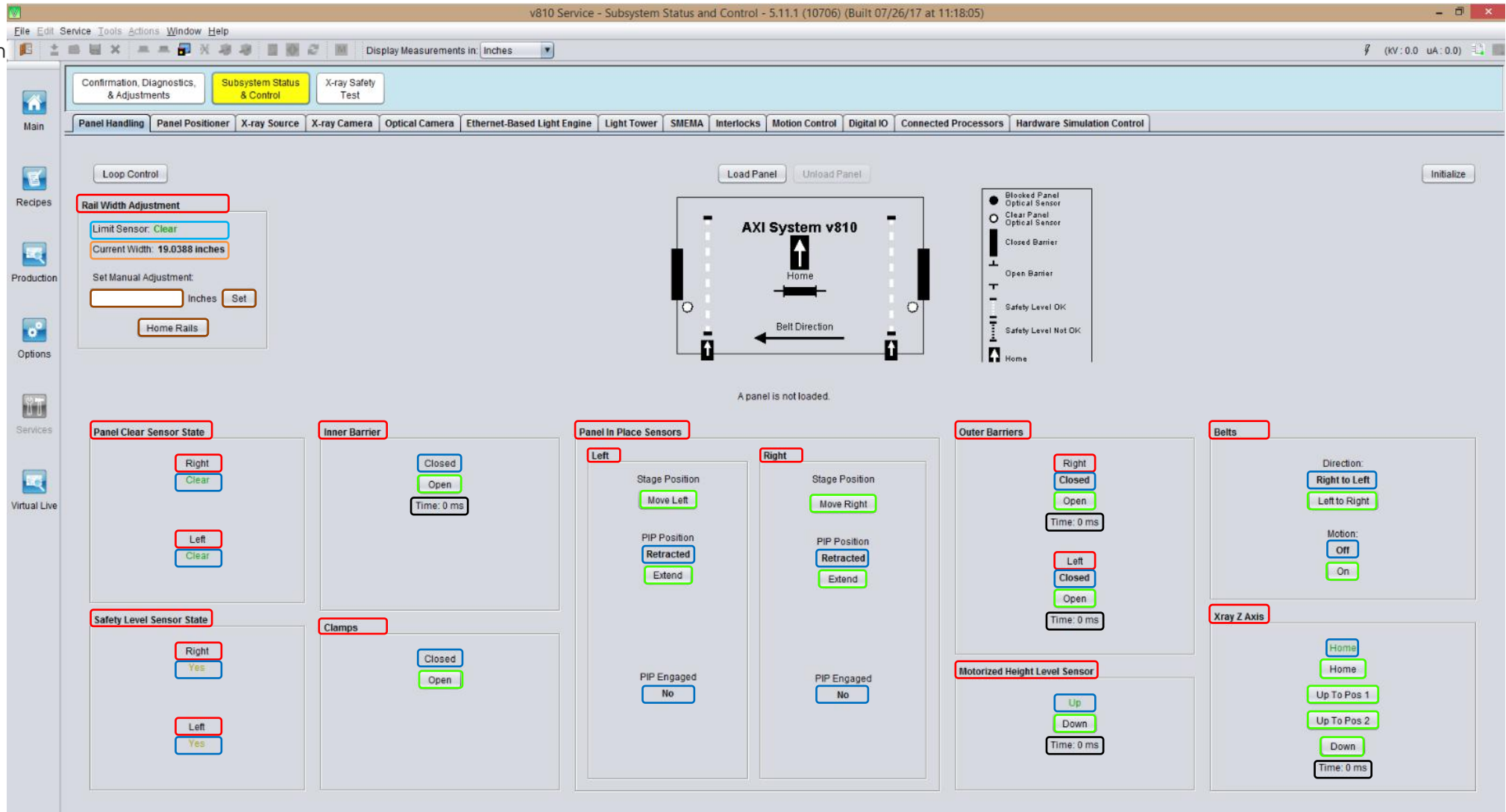
Press button to change state

☐ Left Outer Barrier Open	☐ Right Outer Barrier Open	☐ Enable Optical Stop Controller	☐ Left Panel In Place Sensor Extended
☐ Right Panel In Place Sensor Extended	☐ Reset Optical PIP Sensor	☐ Stage Belts On	☒ Stage Belts Direction Right To Left
☒ Panel Clamps Closed	☐ Panel Clamps Opened	☒ Inner Barrier Closed	☐ Light Stack Red Light On
☒ Light Stack Yellow Light On	☐ Light Stack Green Light On	☐ Light Stack Buzzer On	☒ Z-Axis Enable Signal On
☐ Z-Axis Position Bit A On	☐ Z-Axis Position Bit B On	☐ X-ray Tester Ready To Accept Panel	☐ Panel From X-ray Tester Available
☒ Panel Test Result Good	☐ Panel From X-ray Tester Not Tested	☐ Motorized Height Level Sensor - Reset Alarm	☒ Motorized Height Level Sensor - Homing
☐ Motorized Height Level Sensor - Start Motion	☐ Motorized Height Level Sensor - Position Bit0	☐ Motorized Height Level Sensor - Position Bit1	☐ Motorized Height Level Sensor - Position Bit2
☐ X-ray Cameras Synthetic Trigger On Bar			

Legend

☒ ON ☐ OFF ☒ Error

Automated Board Inspection



Hardware

Status Indicator
(Digital Input Bit)Toggle
(Digital Output Bit)Duration of
hardware status
change (Speed)AWA Home sensor
status

AWA Rail Width

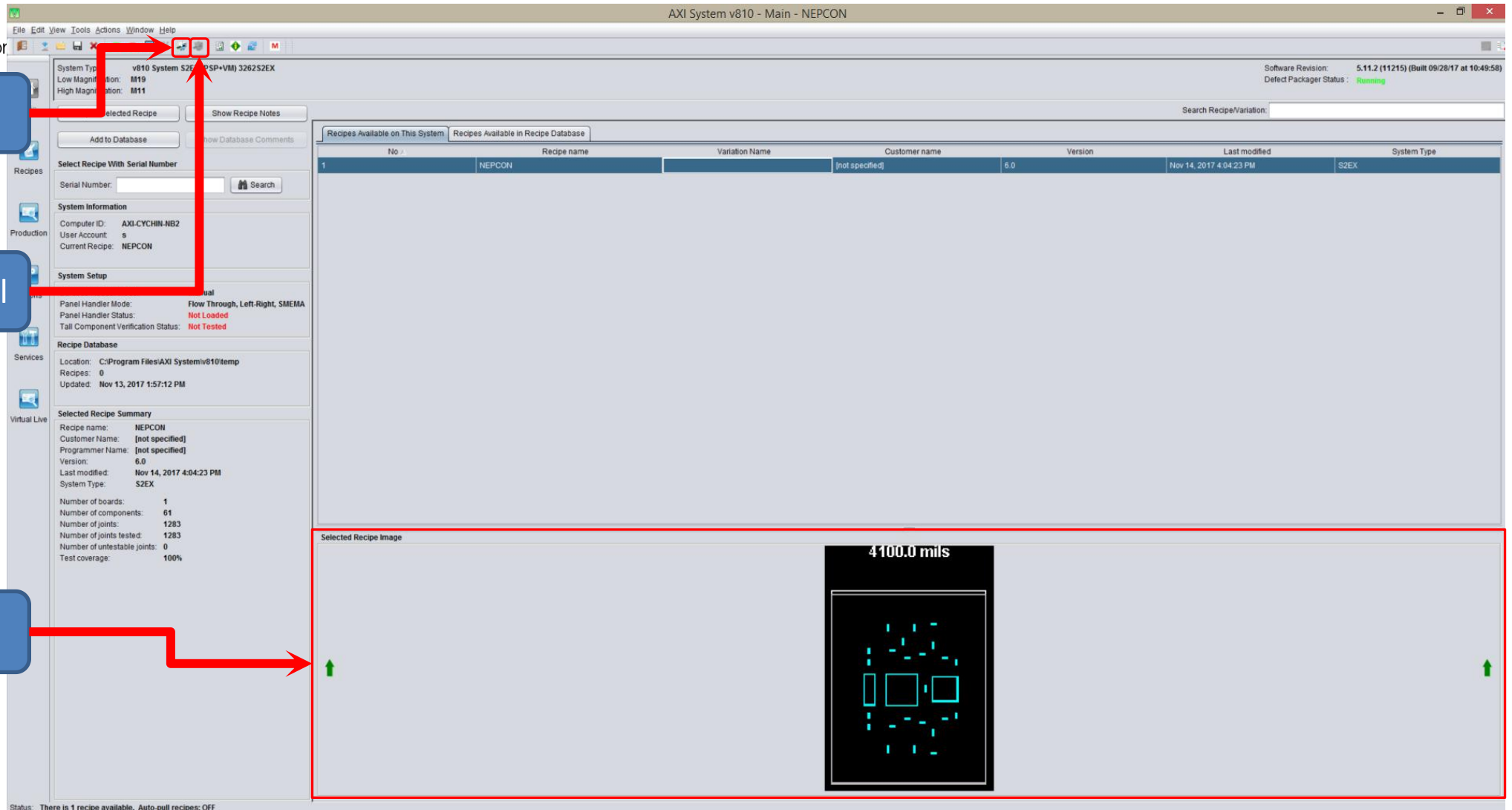
AWA Rail Width
Setting

Automated Board Inspection

Load Panel

Only visible after
select recipes

Unload Panel

Panel load
DirectionIf loading direction
change, image will
change according

Status: There is 1 recipe available. Auto-pull recipes: OFF

v810 Options - Production Options - 5.11.2 (11215) (Built 09/28/17 at 10:49:58) - NEPCON

File Edit Tools Actions Window Help

Software Options Production Options Results Processing Customer Options

Main Recipes Production Options Services Virtual Live

General Serial Number Handling Barcode Reader False Call Triggering

Production Options

- ☒ Eject panel if alignment fails
- ☐ Create result if alignment fails
- ☐ Demo mode
- ☐ Bypass mode
 - ☐ Enable serial number and recipe control during bypass mode
- ☐ Create results during bypass mode
- ☐ Semi-automated mode
- ☒ Enable system light tower buzzer
- ☐ Always Display Variation Name In Repair Station
- ☐ Display Production Panel Loading and Unloading Time
- ☐ Enable X-out detection
 - ☒ Use Alignment failure as X-out Detection. (Must use individual alignment board Method)
 - ☐ Number of failed joints is greater than % of total number of inspected joints on each board.

Confirmation and Adjustment mode:

Script Setup

- ☐ Run program before panel inspection
Path:
- ☐ Run program after panel inspection
Path:
- ☐ Run program for skipped panel

☒ Panel Based ☐ Board Based

Pre Inspection Customize Input Key

Sequence No.	Parameter
☐	eqpCode
☐	projectName
☐	userName
☐	panelWidthInNanometer

Pre Inspection Customize Output Key

Sequence No.	Parameter
☐	warning
☐	error
☐	panelSerialNumber

Sampling Options

- ☐ Panel sampling
 - ☒ Test 1 in every panels OR ☐ Skip 1 in every panels

Panel Handling

- ☒ Use SMEMA communications
 - ☐ Ignore PIP sensor during board loading
 - ☒ Send "Panel From XrayTester Available signal" to downstream after receive "Downstream Waiting Accept Panel Signal" during unloading
 - ☐ Remain Outer Barrier Open For Loading After Unloading Panel (Only Applicable to Pass Back Loading Mode)
 - ☒ Open Outer Barrier Immediately After Scanning Complete
 - ☐ Check Downstream Sensor Clear Signal During Board Unloading

Loading mode:

Flow direction:

Repair Images Save

- ☐ Save focus confirmation repair data
 - * Checking this box could increase the number of repair images saved by a factor of 5
- ☐ Limit number of repair images saved
 - Max number of repair images to save

Repair Image Type: ☒ JPG ☐ PNG

Status:

Load board direction
and SMEMA enable



GUI Interface - Panel Handling Subsystem - Manual Load Board

Automated Board Inspection

v810 Service - Subsystem Status and Control - 5.11.1 (10706) (Built 07/26/17 at 11:18:05)

File Edit Service Tools Actions Window Help

Display Measurements in **Inches**

Confirmation, Diagnostics, & Adjustments Subsystem Status & Control X-ray Safety Test

Panel Handling Panel Positioner X-ray Source X-ray Camera Optical Camera Ethernet-Based Light Engine Light Tower SMEMA Interlocks Motion Control Digital IO Connected Processors Hardware Simulation Control

Loop Control

Rail Width Adjustment

Limit Sensor: **Clear**

Current Width: 19.0388 inches

Set Manual Adjustment:

inches **Set**

Home Rails

AXI System v810

Home

Load Panel

Unload Panel

Initialize

Maximum panel size (inches): 19' x 24'

Minimum panel size (inches): 3' x 3'

Panel Clear Sensor State

Right **Clear**

Left **Clear**

Safety Level Sensor State

Right **Yes**

Left **Yes**

Inner Barrier

Closed

Open

Time: 0 ms

Panel In Position

Left

Move Left

PIP Position Retracted

Extend

PIP Engaged No

Right

Move Right

PIP Position Retracted

Extend

PIP Engaged No

Outer Barriers

Right Closed

Open

Belts

Direction: Right to Left

Left to Right

Motion: Off

On

Xray Z Axis

Home

Up To Pos 1

Up To Pos 2

Down

Time: 0 ms

Motorized Height Level Sensor

Up

Down

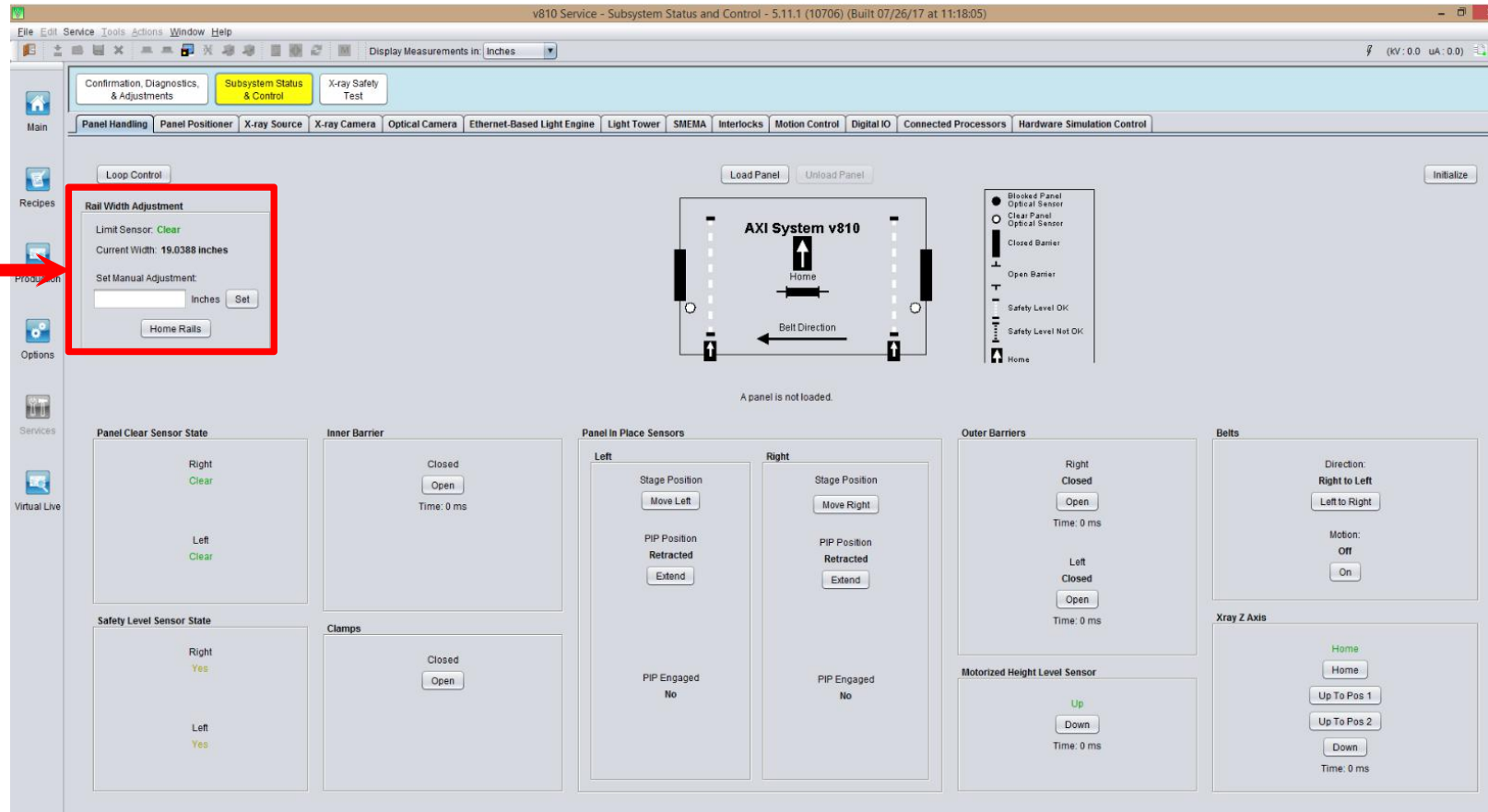
Time: 0 ms



Automated Board Inspection

GUI Interface - Panel Handling Subsystem - AWA Rail Width

AWA Rail Width self define



- AWA rail width home position and maximum rail width is 19 inches
- AWA minimum rail width is 3 inches
- AWA Limit sensor should always show “Clear”
- Manual set rail width value should same as physical measurement
- Panel loaded with toggle “Load Panel” will automatically + 0.3 mm as buffer to avoid incoming PCBA panel over width and board stuck at AWA

ABI Hardware Overview - Panel Handling

Automated Board Inspection

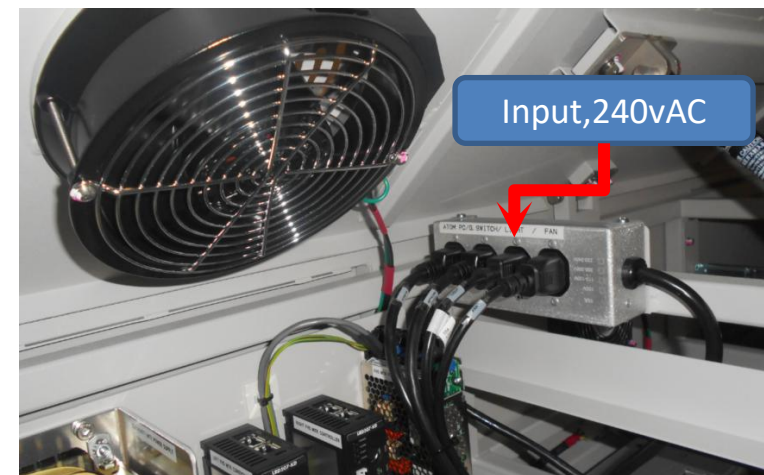
- Stage service lamp (Manual On/Off)
- Inner Barrier
- Outer Barriers
- XY Stage
 - Auto Width Adjustment (AWA)
 - Panel Clear Sensors
 - Belts, Motors, & Relays
 - Panel In Place Sensor (PIP)
 - Panel Clamps
- Motion Controller and Digital IO Assembly



Left Service Lamp



Right Service Lamp



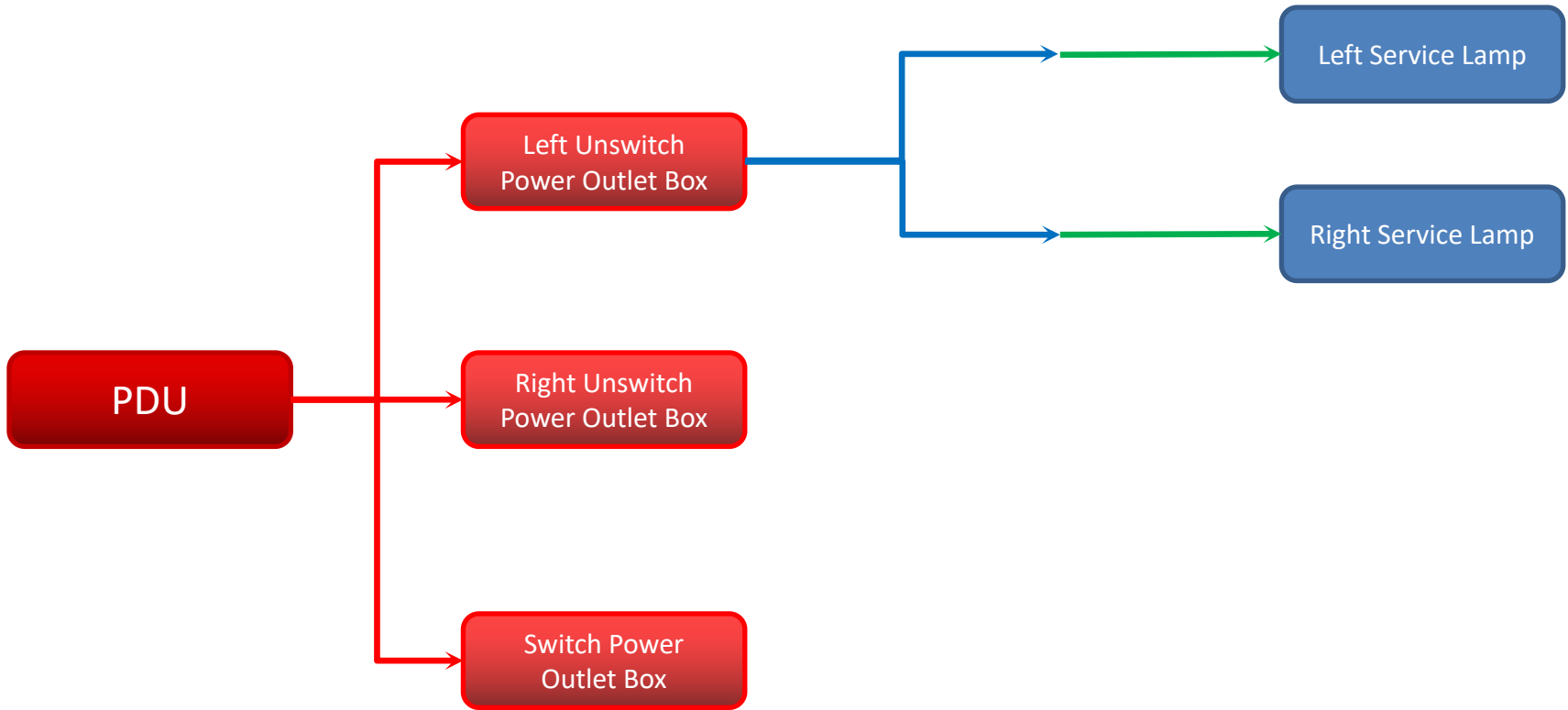
Left Server Rack, Unswitch Power Outlet Box

- Service lamp used is LED type, 9W, 570MM length, COOL WHITE
- Direct input from Unswitch Power Outlet Box, 240vAC
- Manual On/Off button



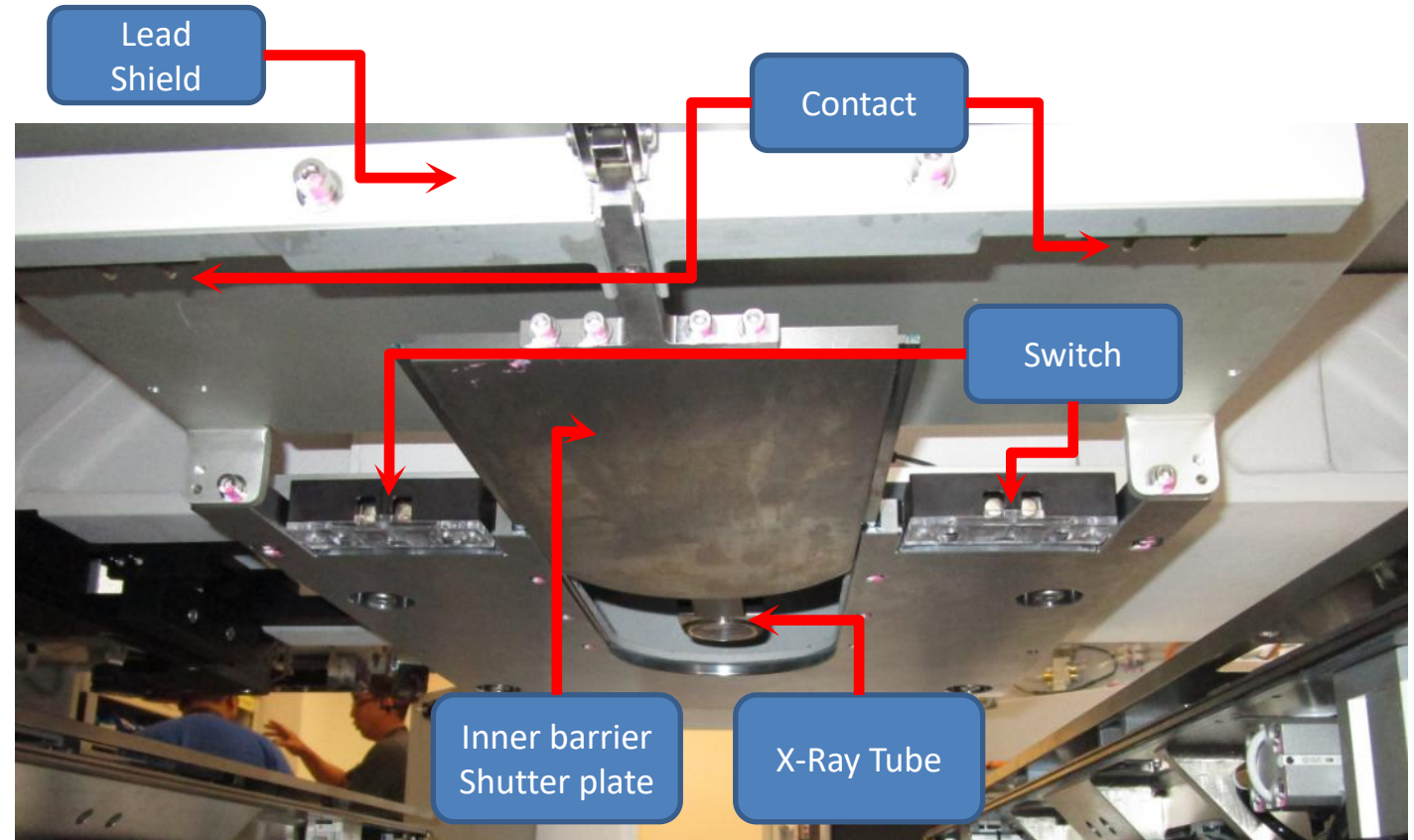
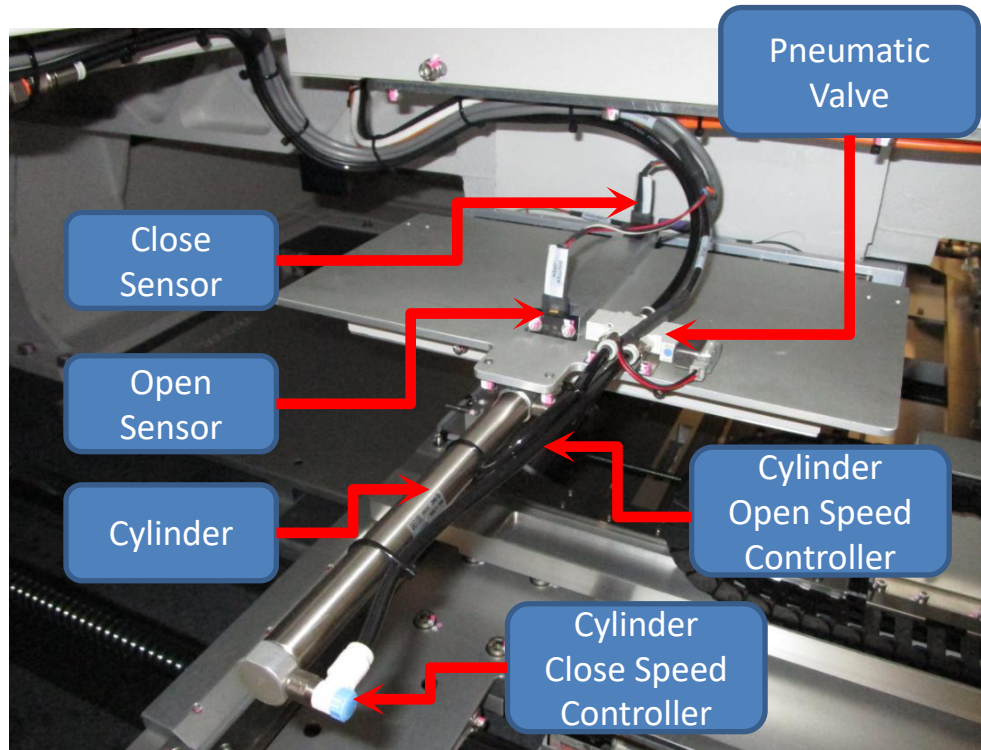
Automated Board Inspection

Hardware Overview - Stage View Service Lamp Connectivity



ABI Hardware Overview - Inner Barrier

Automated Board Inspection



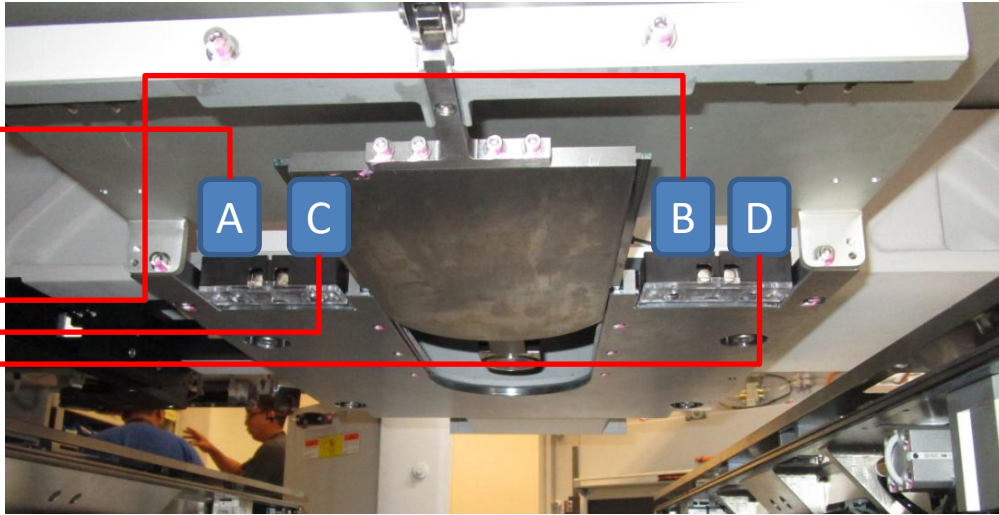
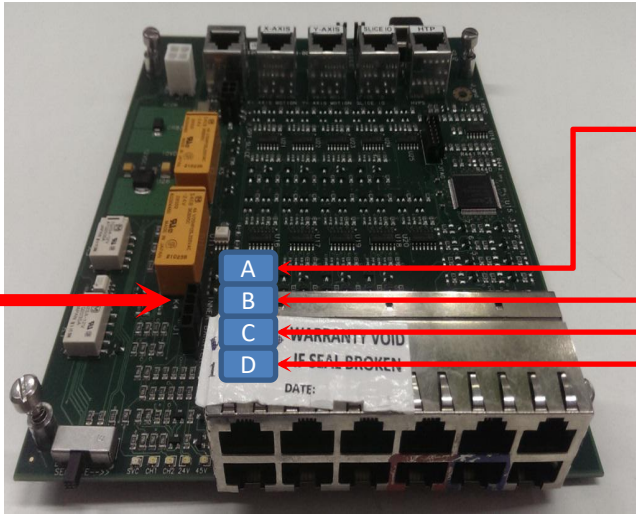
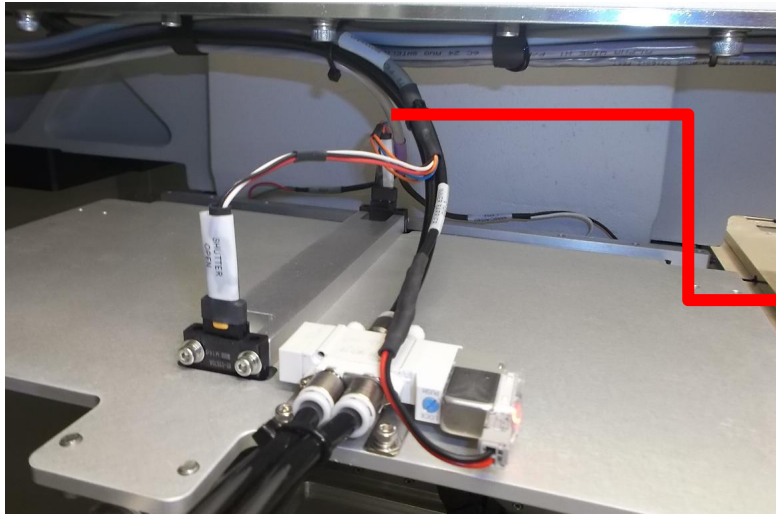
➤ Opens to expose the panel to X-Rays for imaging.

➤ Closes to contain X-Rays inside for loading and unloading the panel.

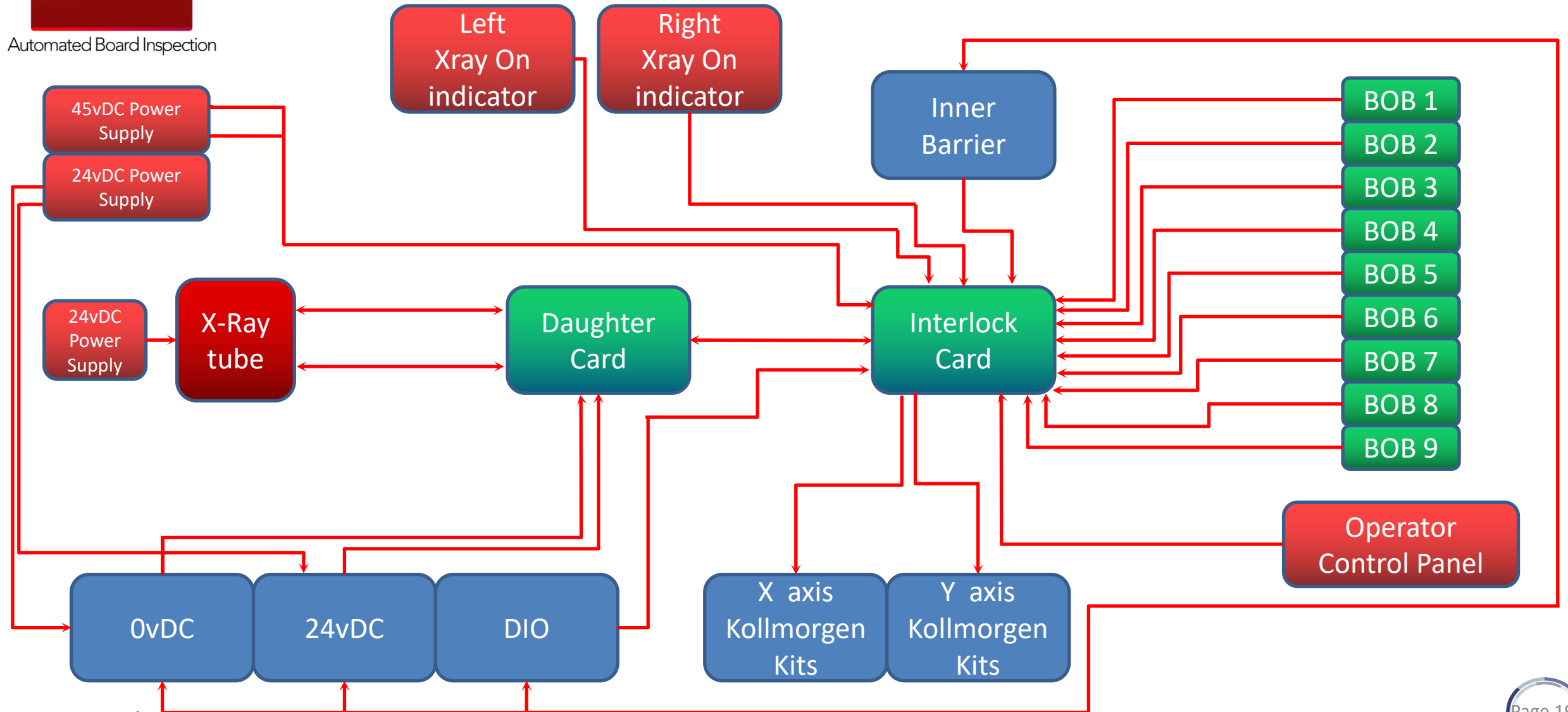


Hardware Overview - Interlock Connectivity

Automated Board Inspection



Automated Board Inspection



ABI Hardware Overview - Inner Barrier - Interlock Card

Automated Board Inspection

Connect to Daughter Card

Connect to DIO Slide-4

Connect to Y Axis kollomorgen

Connect to X Axis kollomorgen

Connect to 45vDC & 24 vDC

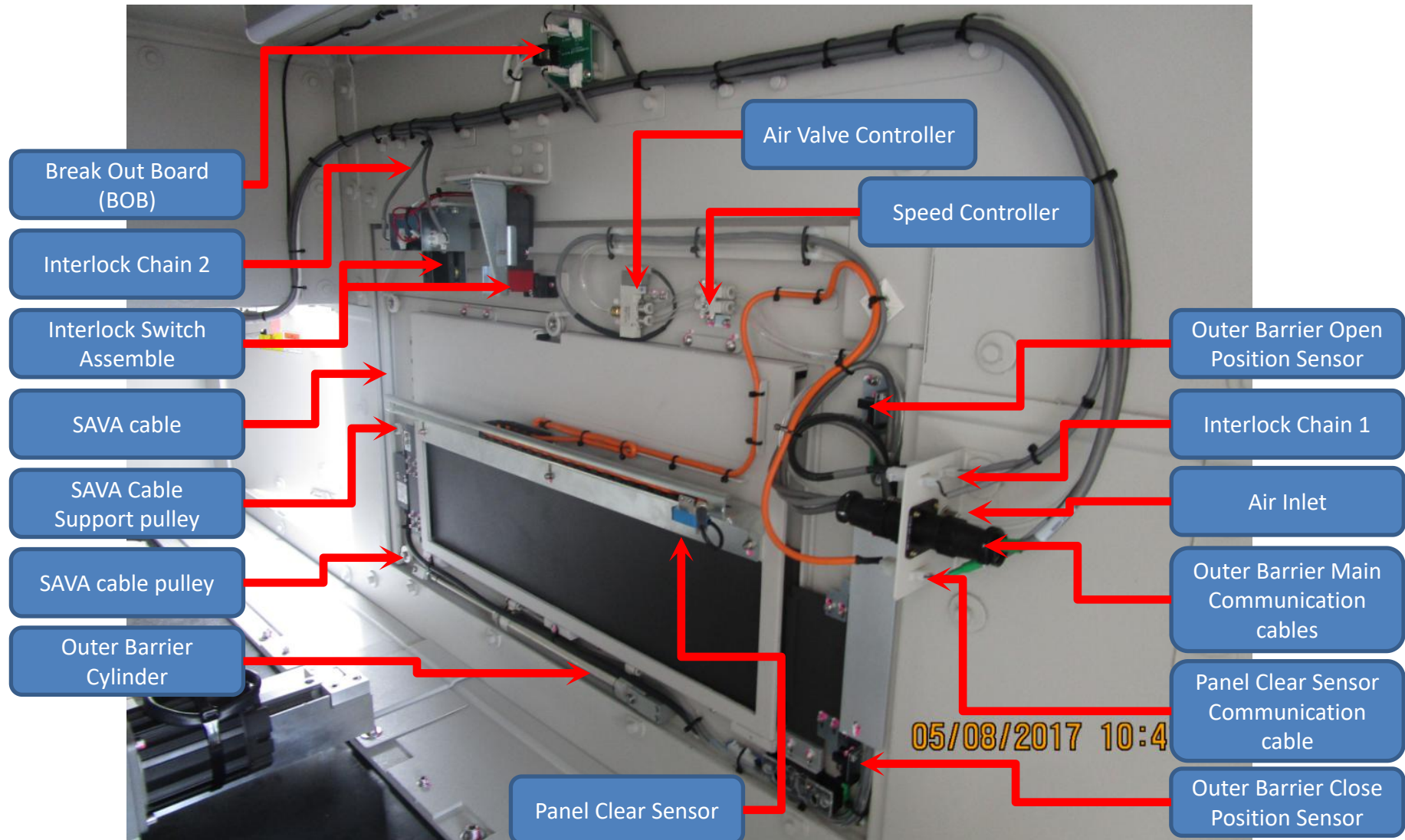
Connect to DIO Slide-4

Top

Connect to Inner Barrier



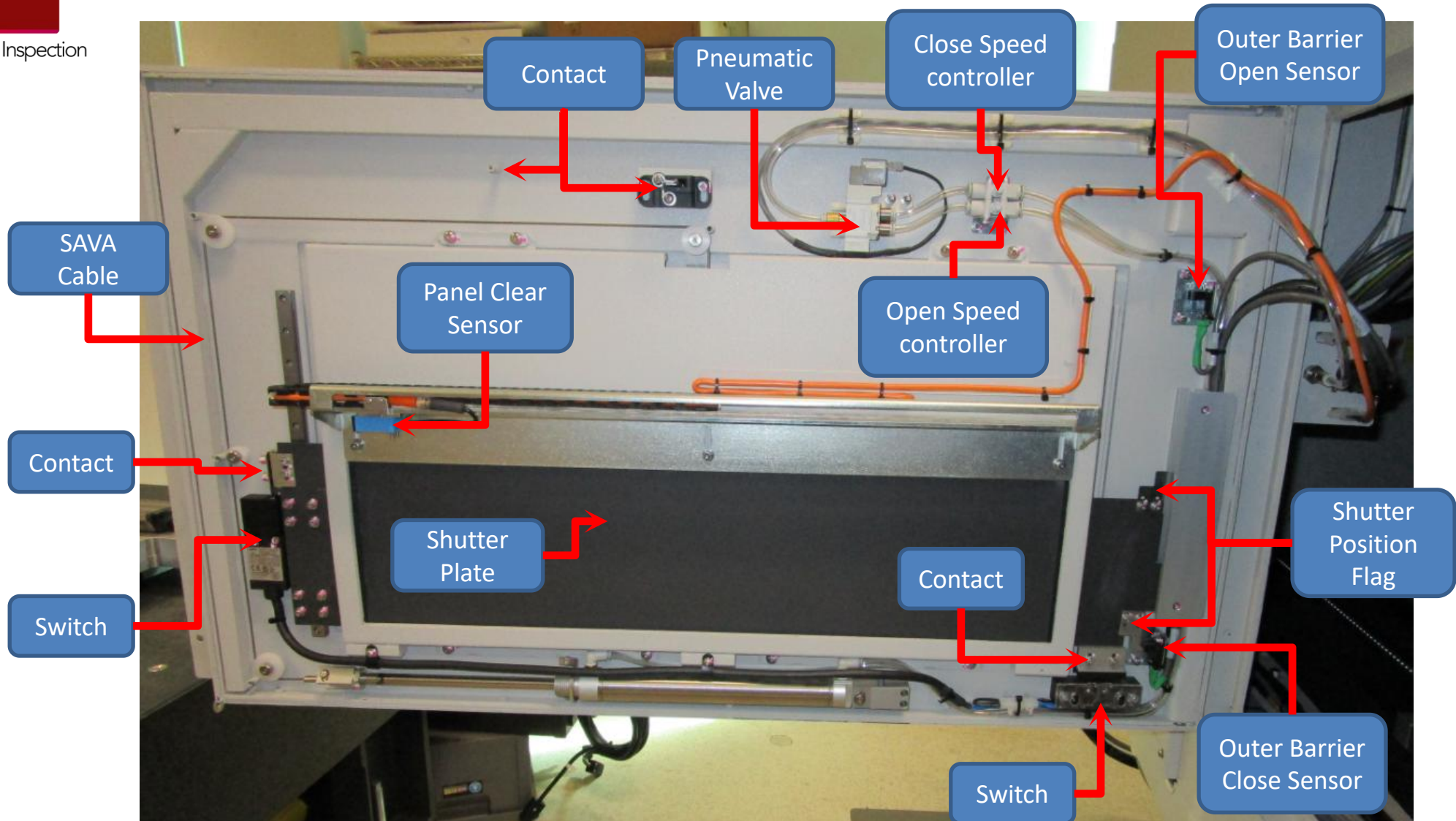
Bottom



➤ Outer barrier open in order to load the board inside the machine

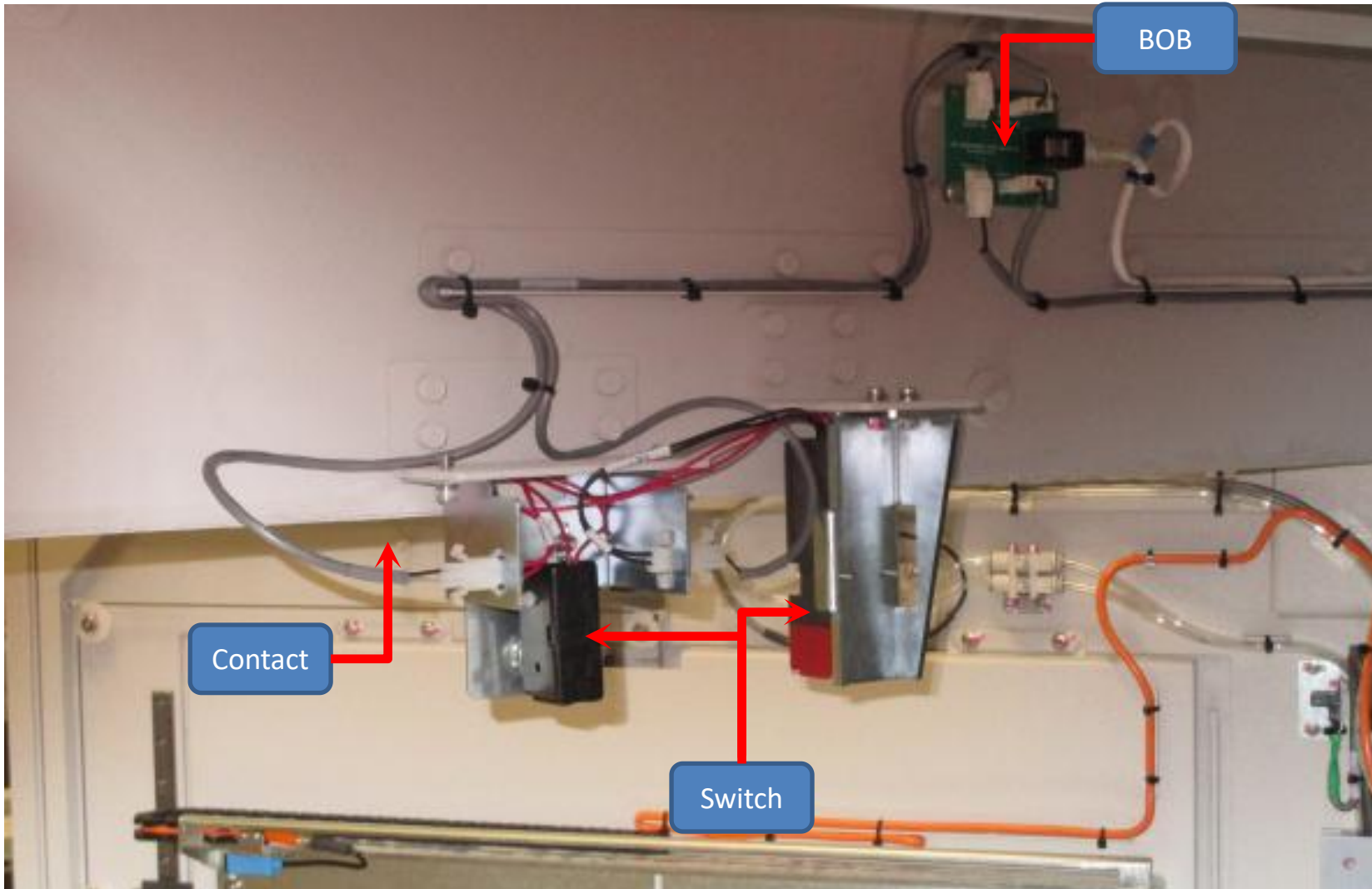
ABI Hardware Overview - Outer Barrier

Automated Board Inspection



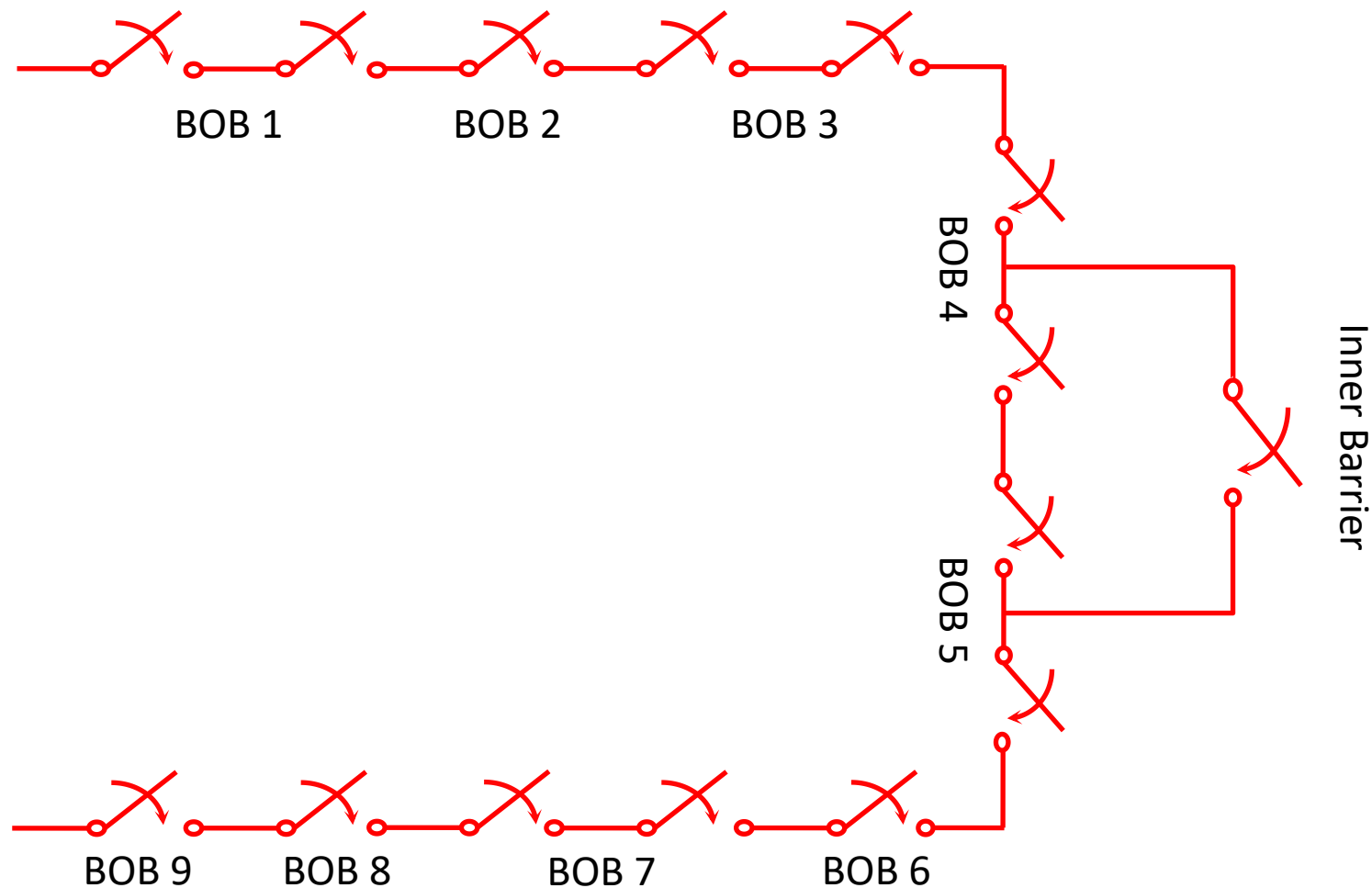
ABI Hardware Overview - Outer Barrier

Automated Board Inspection

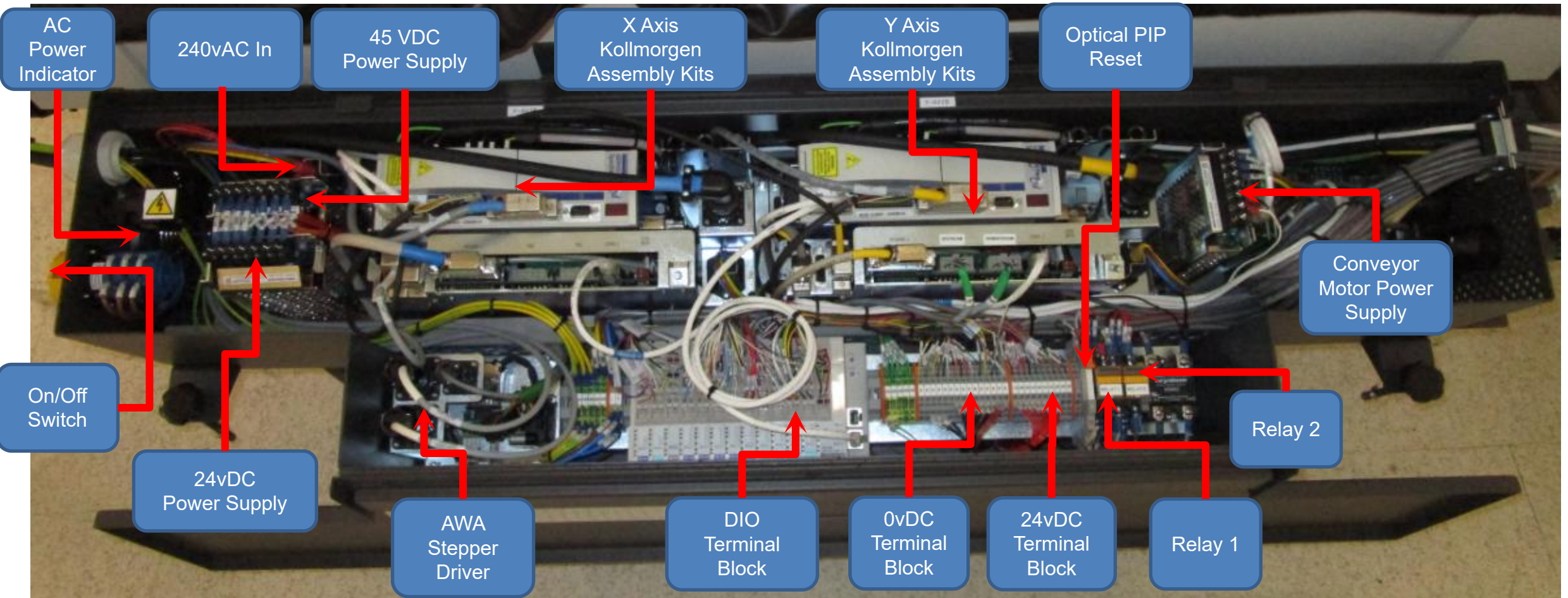


ABI Interlock Subsystem Working Concept

Automated Board Inspection



Automated Board Inspection



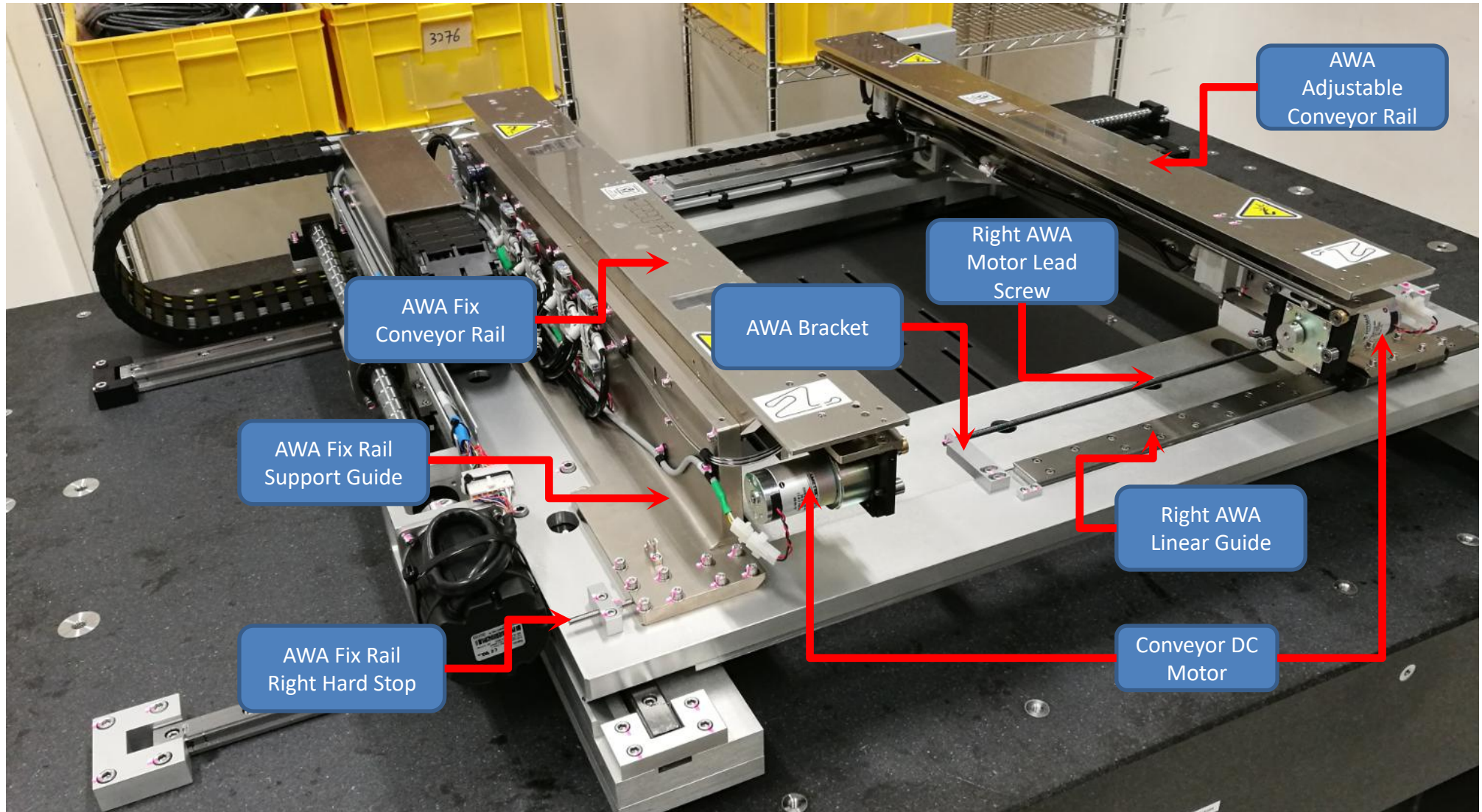
ABI Hardware Overview - XY Stage

Automated Board Inspection



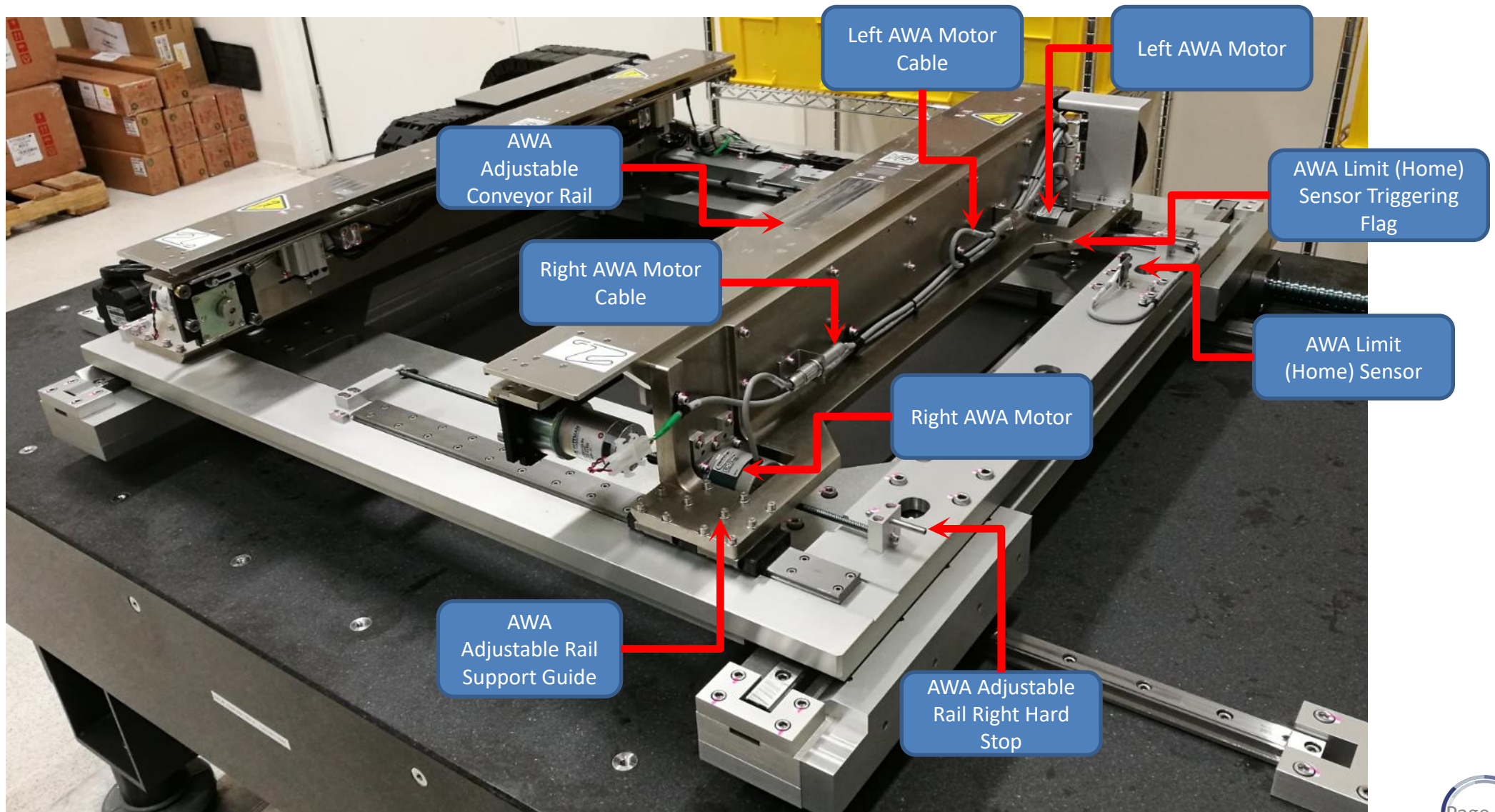
ABI Hardware Overview - Panel Handling - XY Stage & AWA

Automated Board Inspection



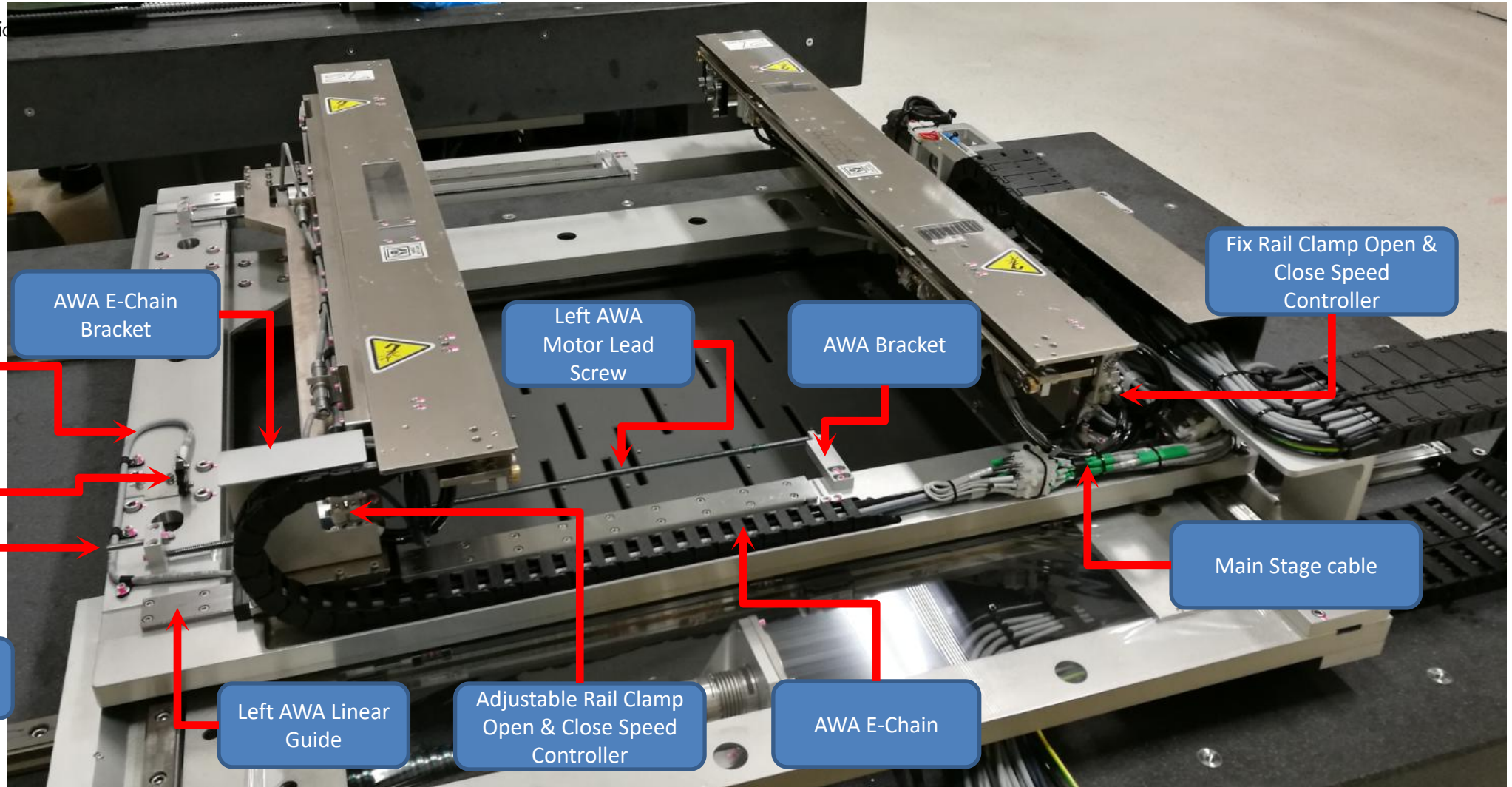
ABI Hardware Overview - Panel Handling - XY Stage & AWA

Automated Board Inspection



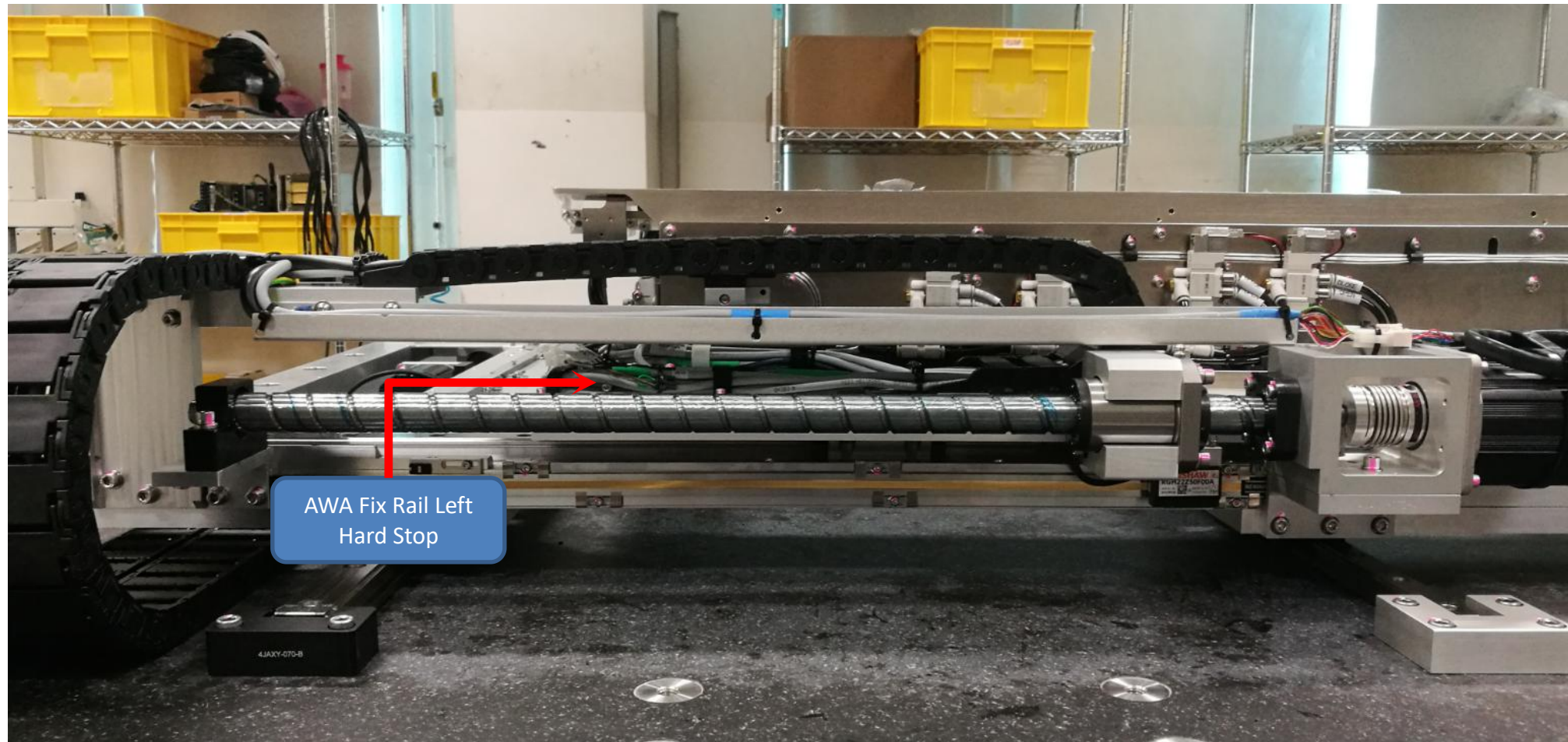
ABI Hardware Overview - Panel Handling - XY Stage & AWA

Automated Board Inspection

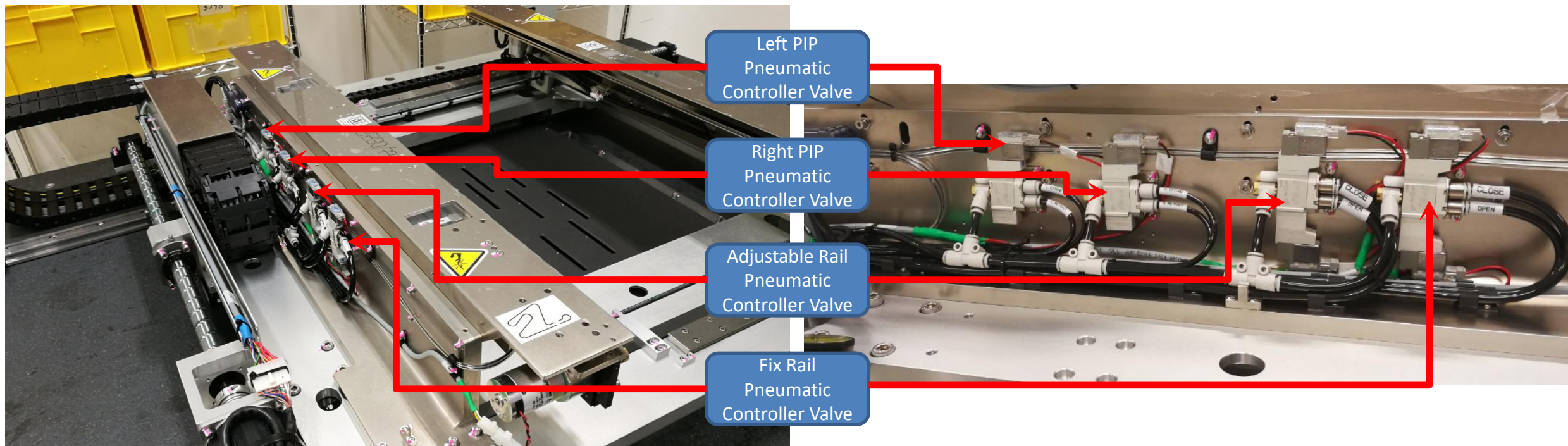


ABI Hardware Overview - Panel Handling - XY Stage & AWA

Automated Board Inspection



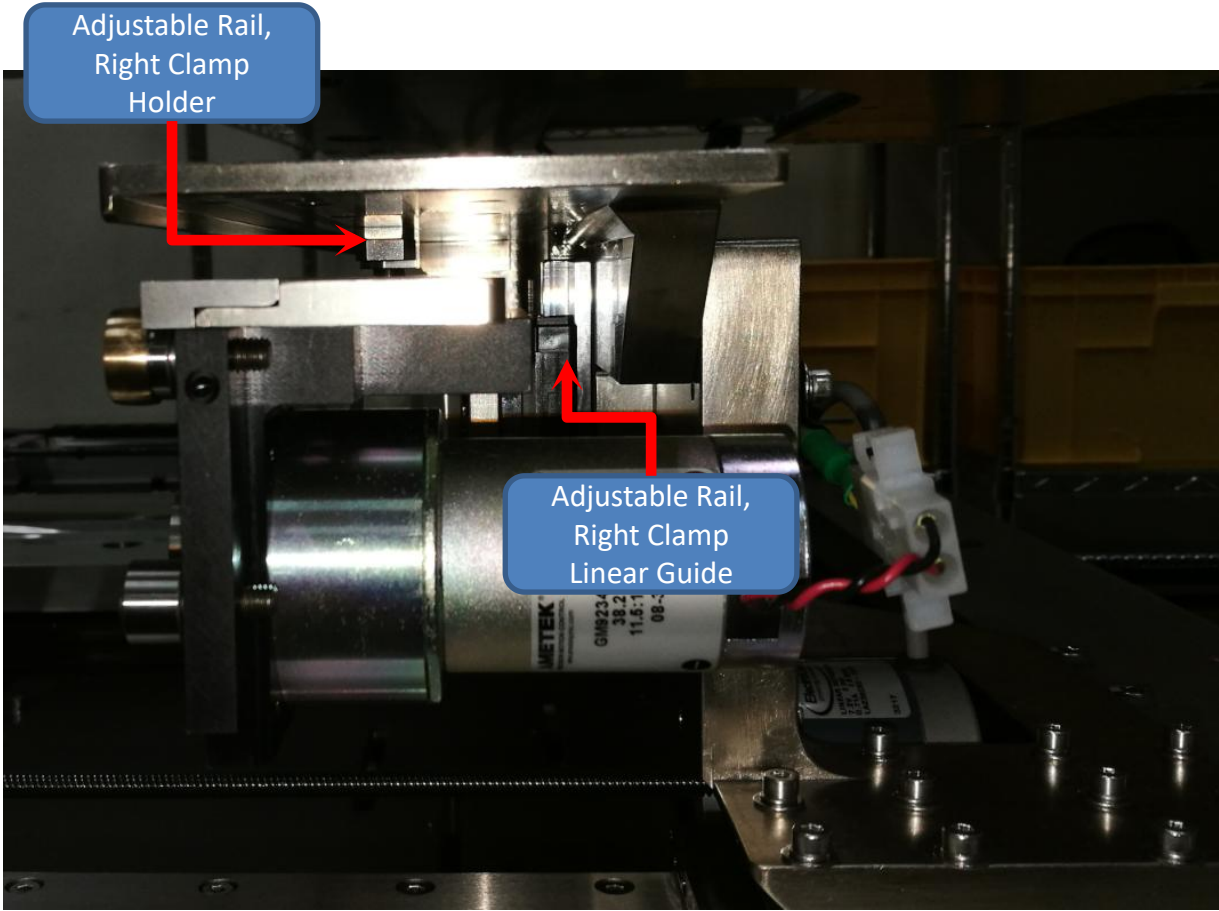
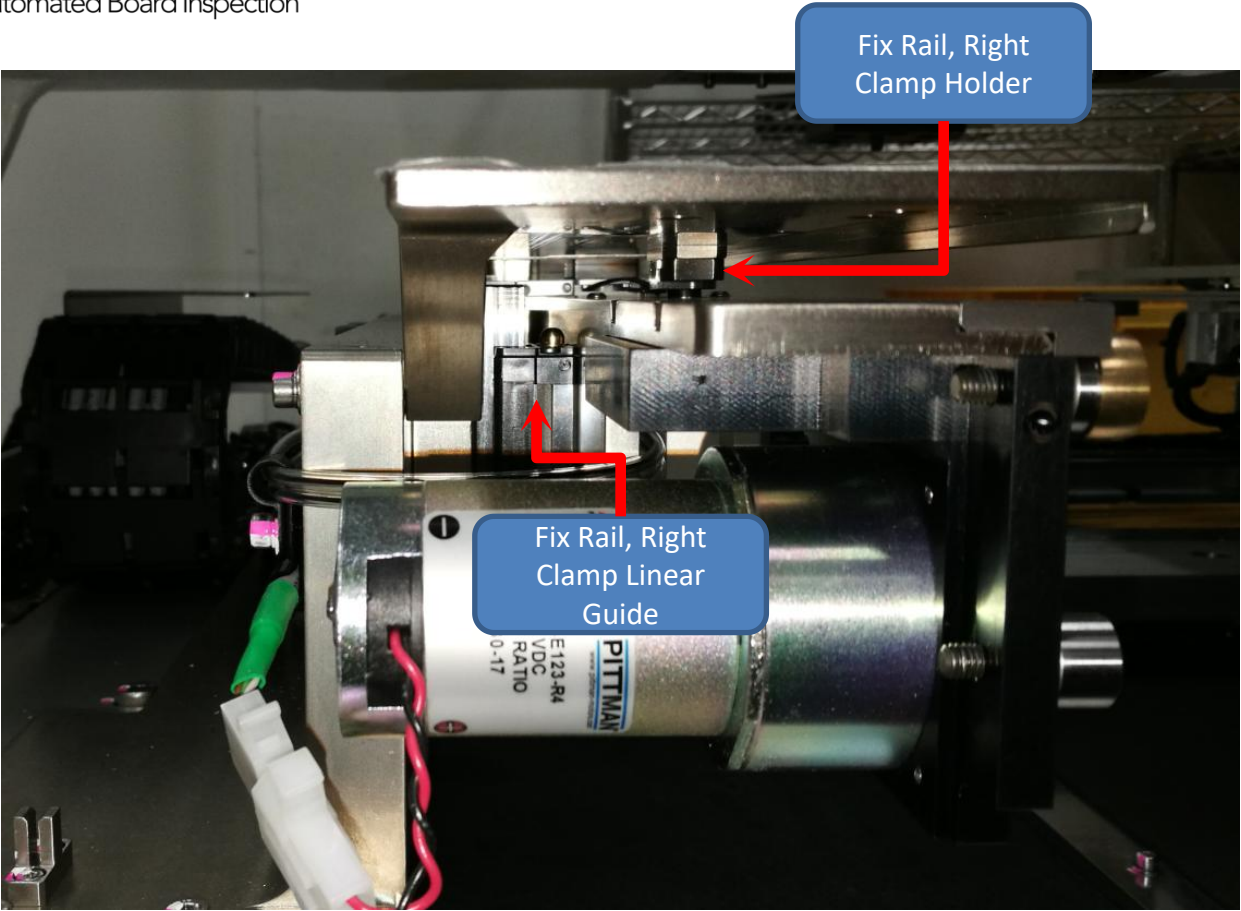
AWA Fix Rail Left
Hard Stop





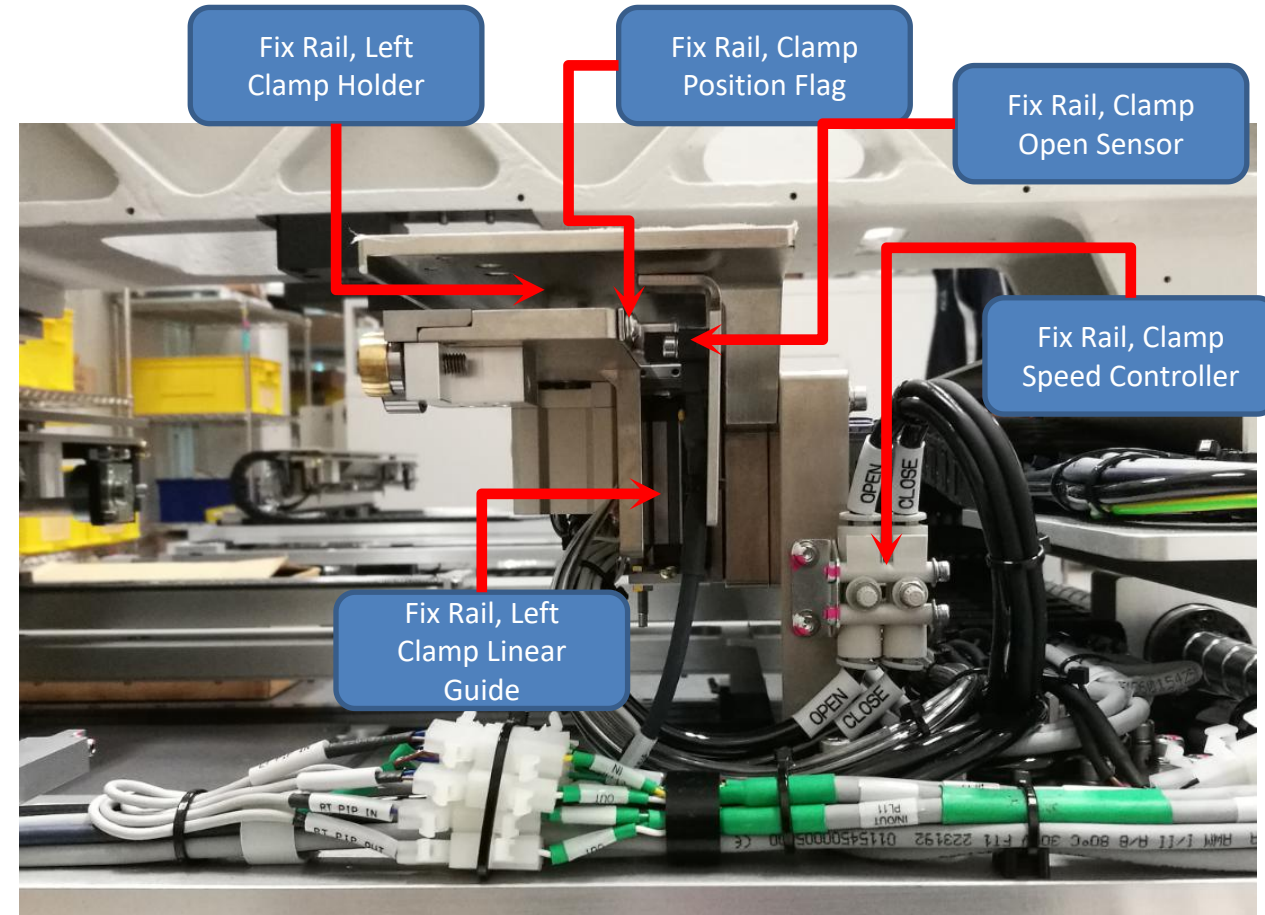
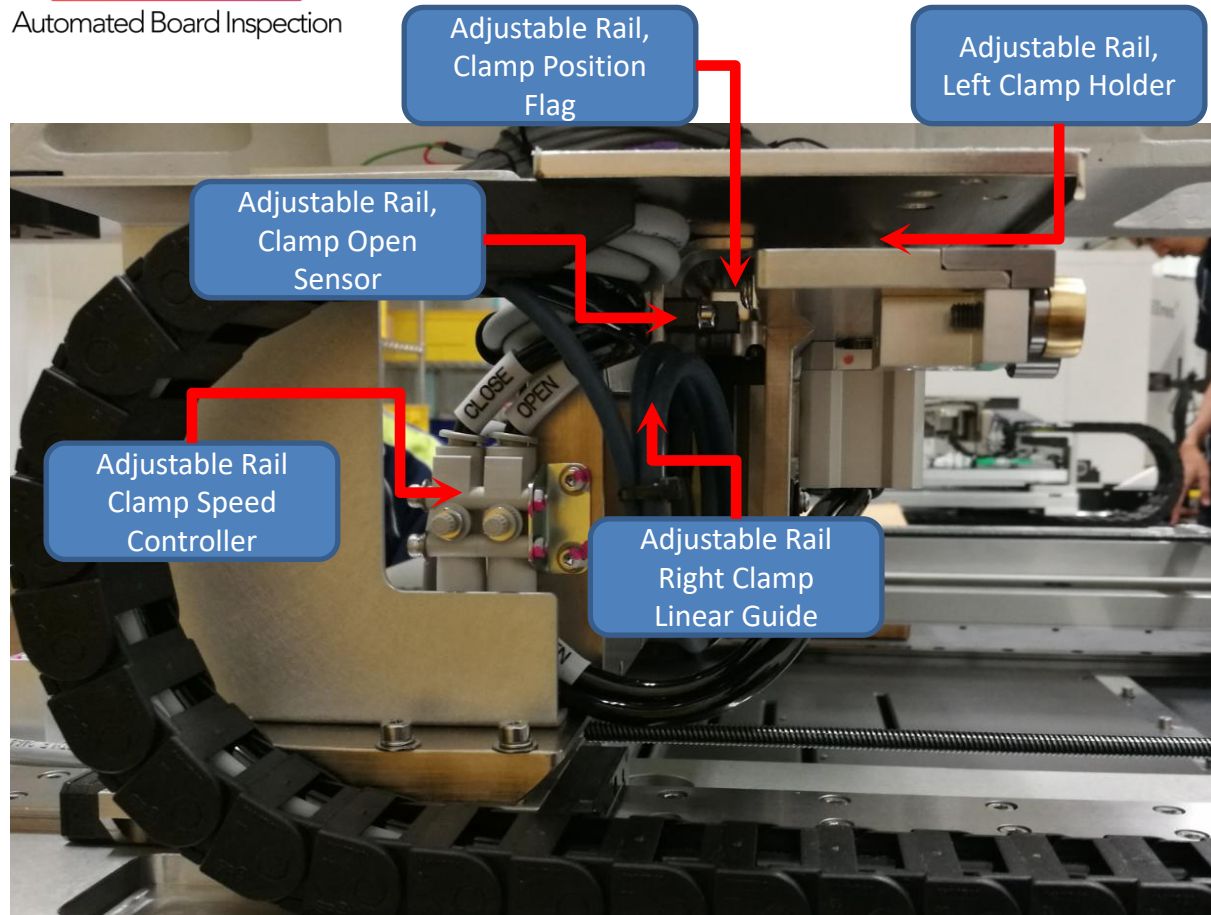
Hardware Overview - Panel Handling - Panel Clamp

Automated Board Inspection



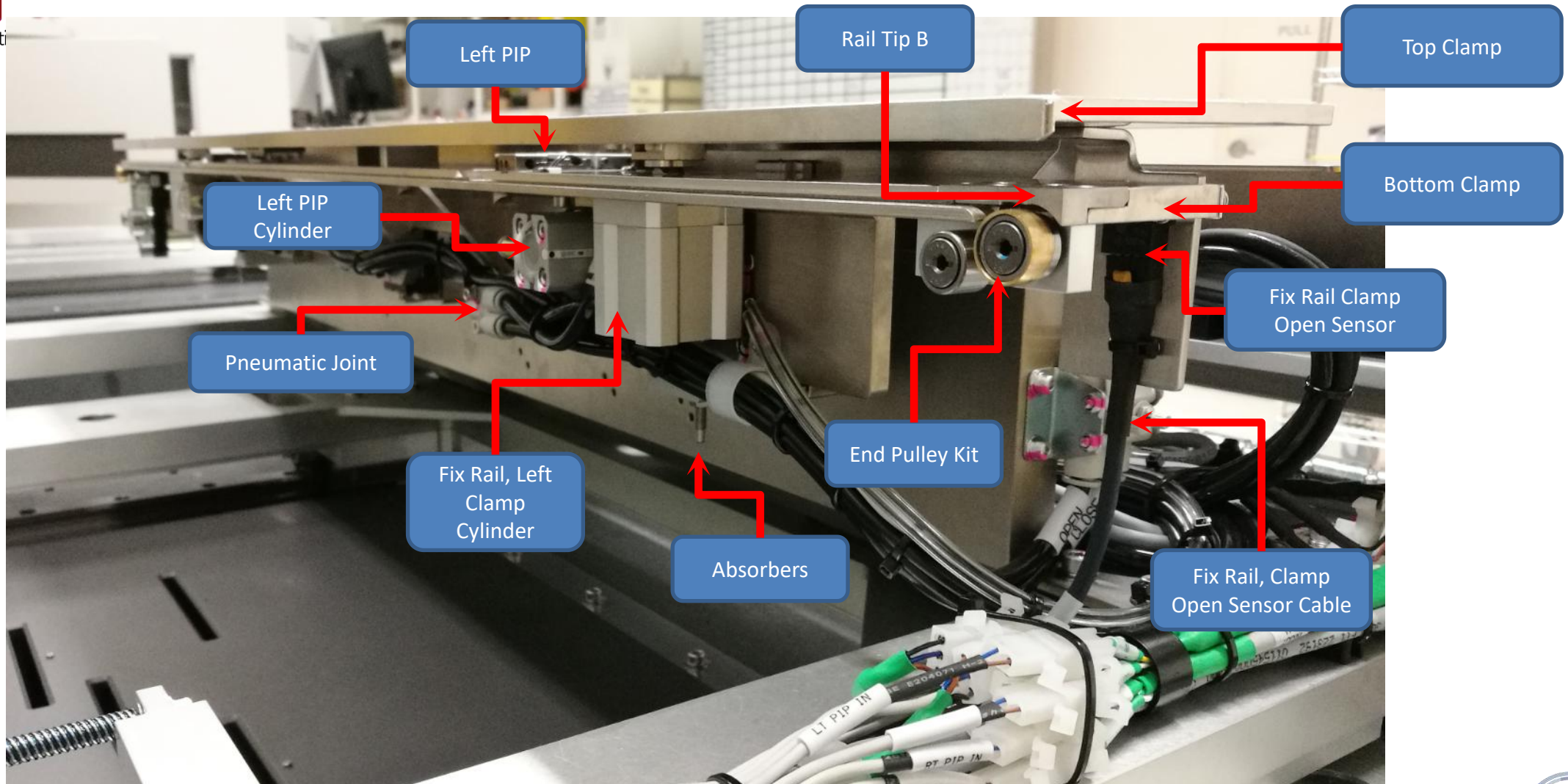
ABI Hardware Overview - Panel Handling - Panel Clamp

Automated Board Inspection



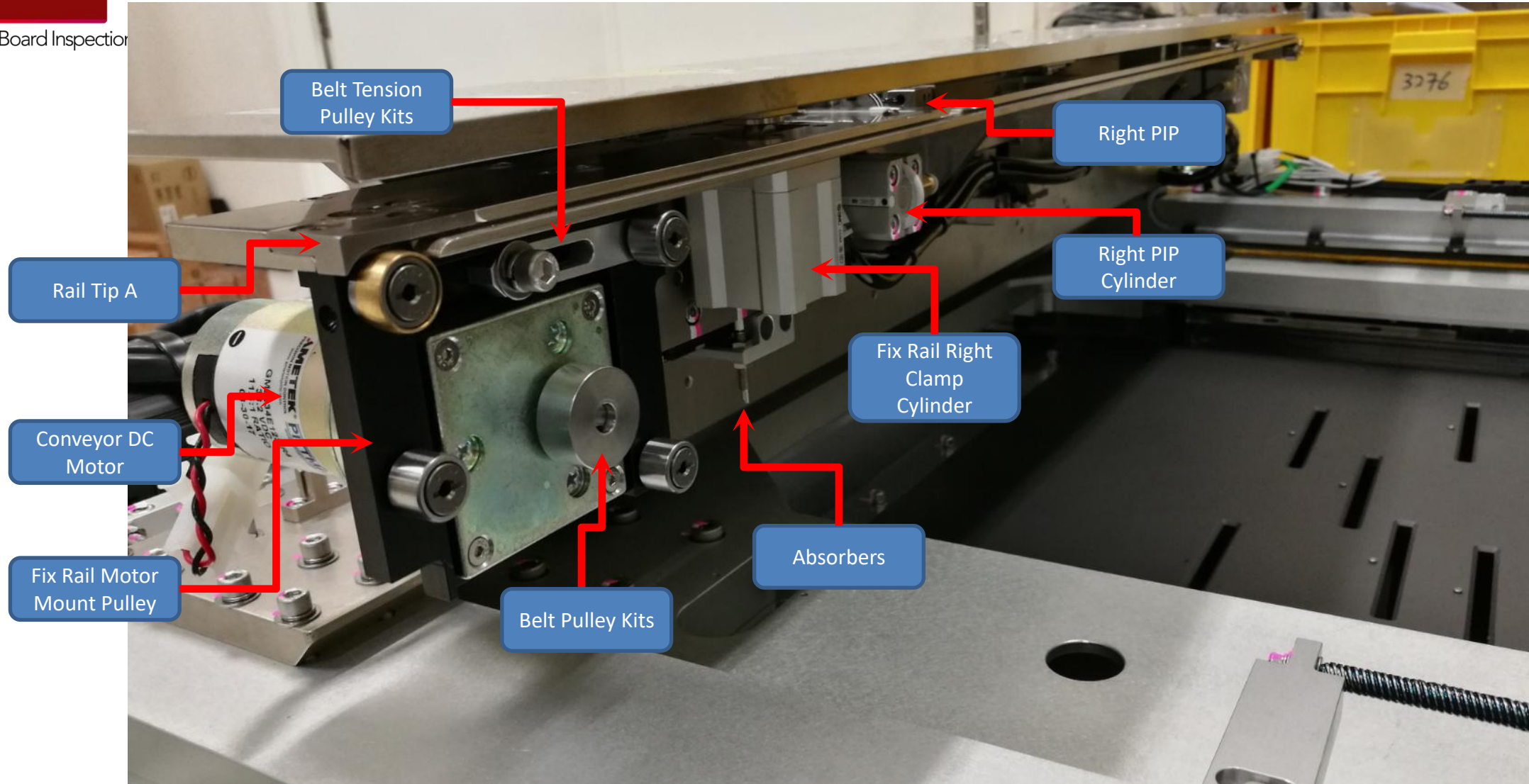
ABI Hardware Overview - XY Stage - Fix Rail

Automated Board Inspection



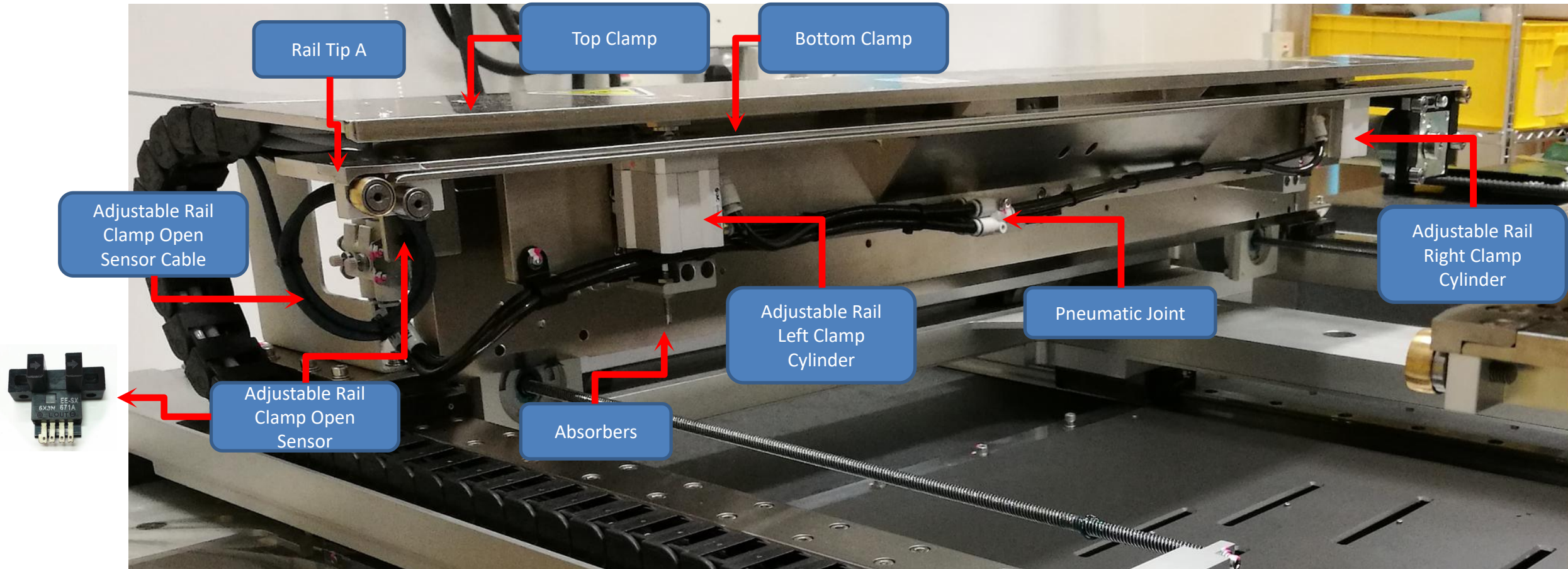
ABI Hardware Overview - XY Stage - Fix Rail

Automated Board Inspection



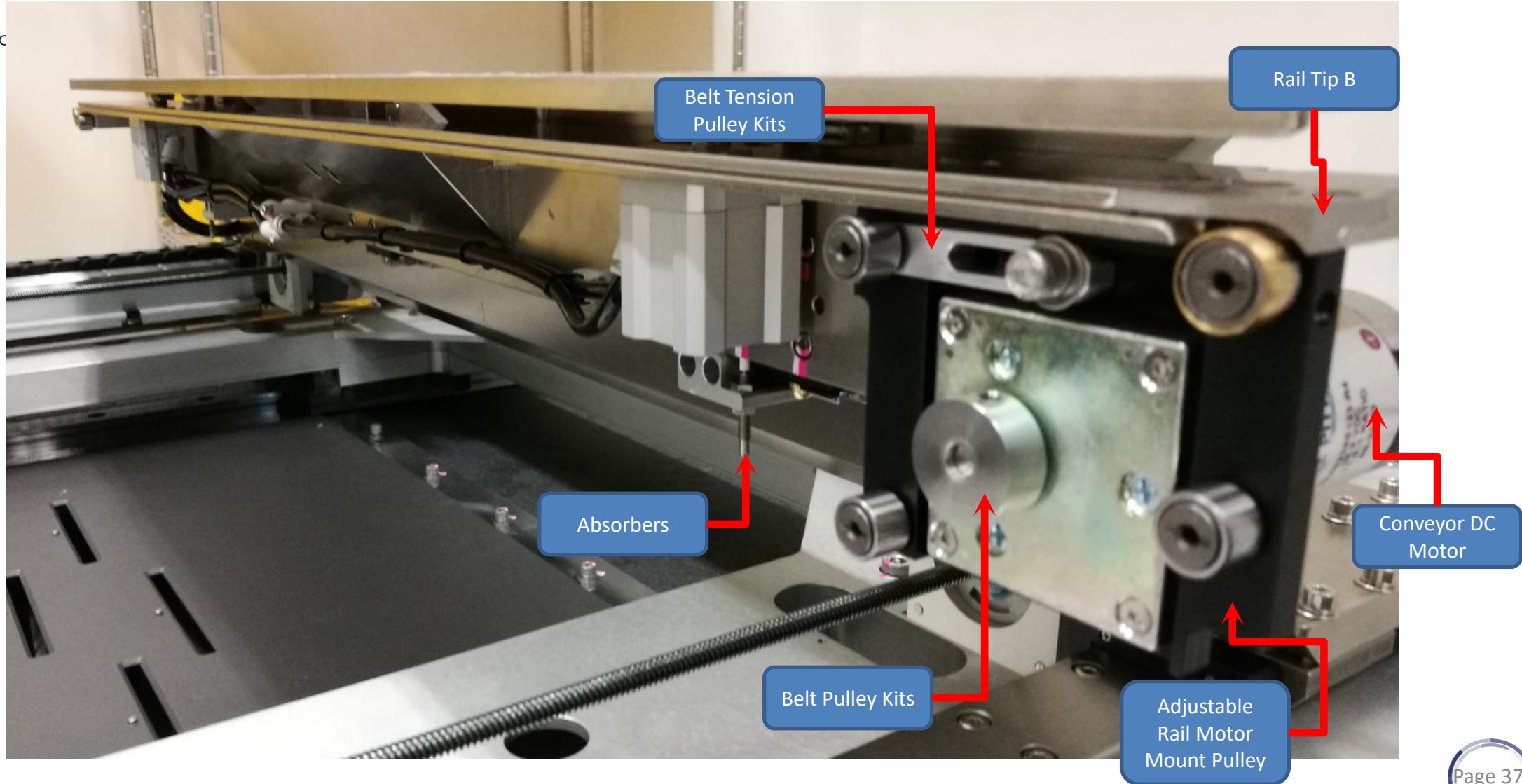
ABI Hardware Overview - XY Stage - Adjustable Rail

Automated Board Inspection



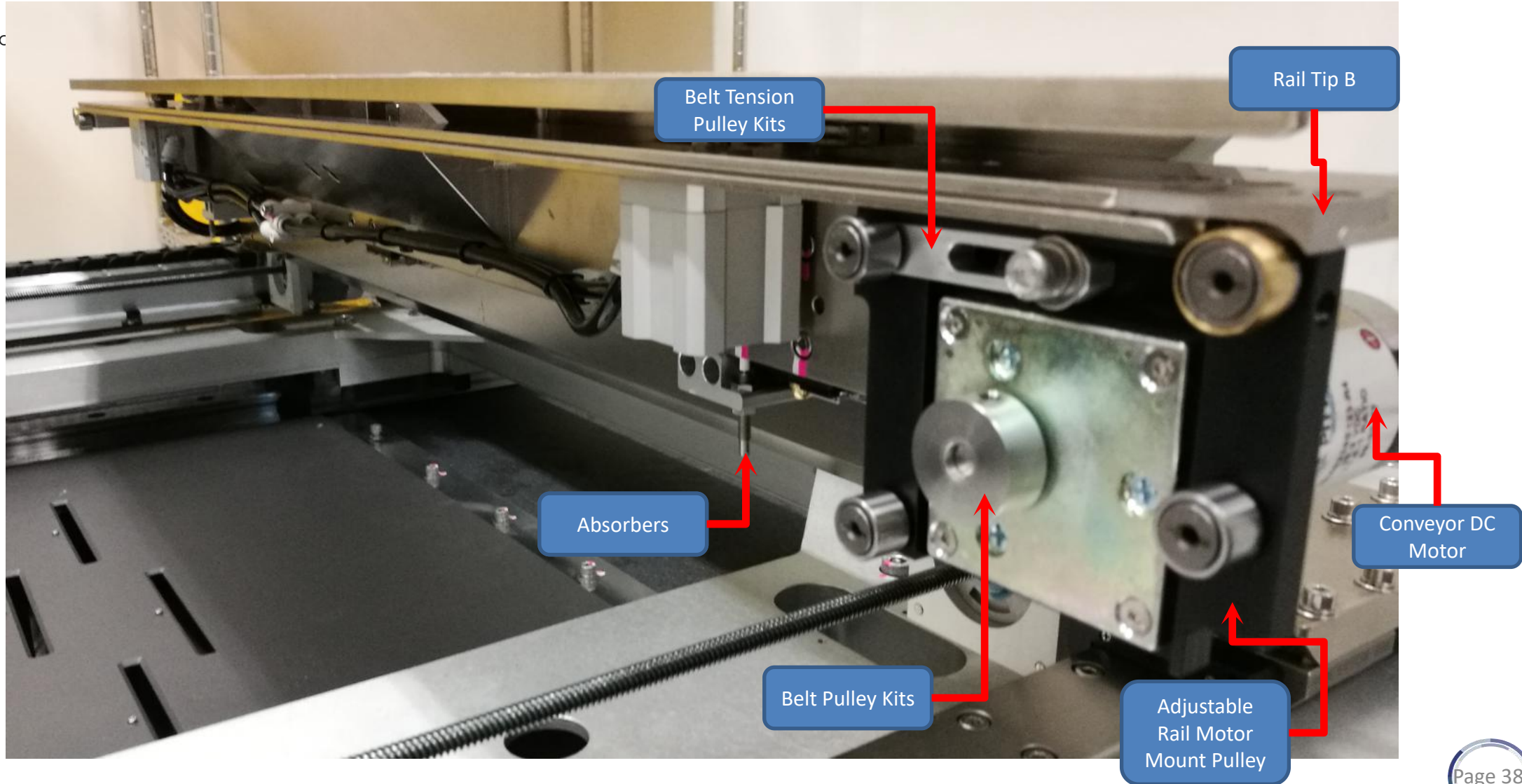
ABI Hardware Overview - XY Stage - Adjustable Rail

Automated Board Inspection



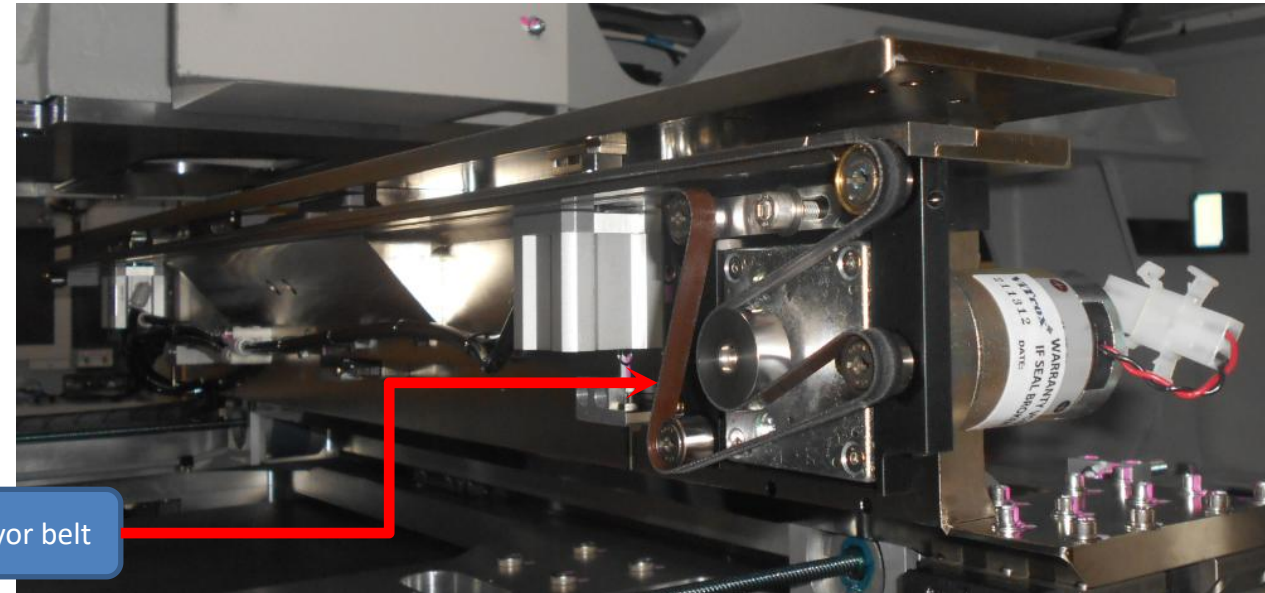
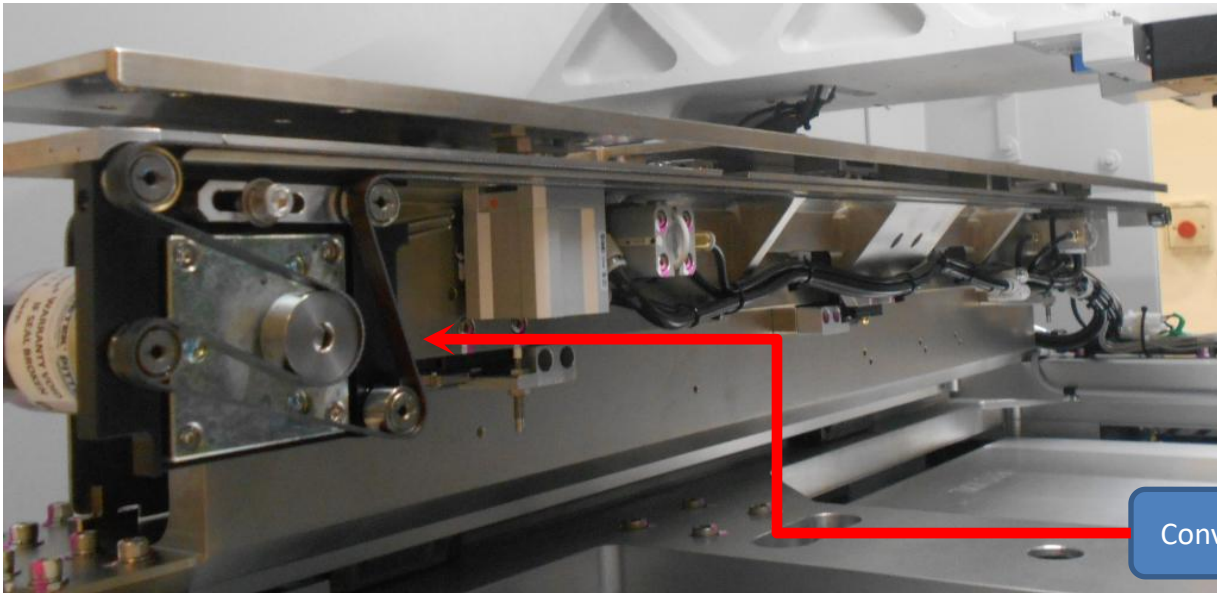
ABI Hardware Overview - XY Stage - Adjustable Rail

Automated Board Inspection



ABI Hardware Overview - XY Stage - Conveyor Belt

Automated Board Inspection



Conveyor belt



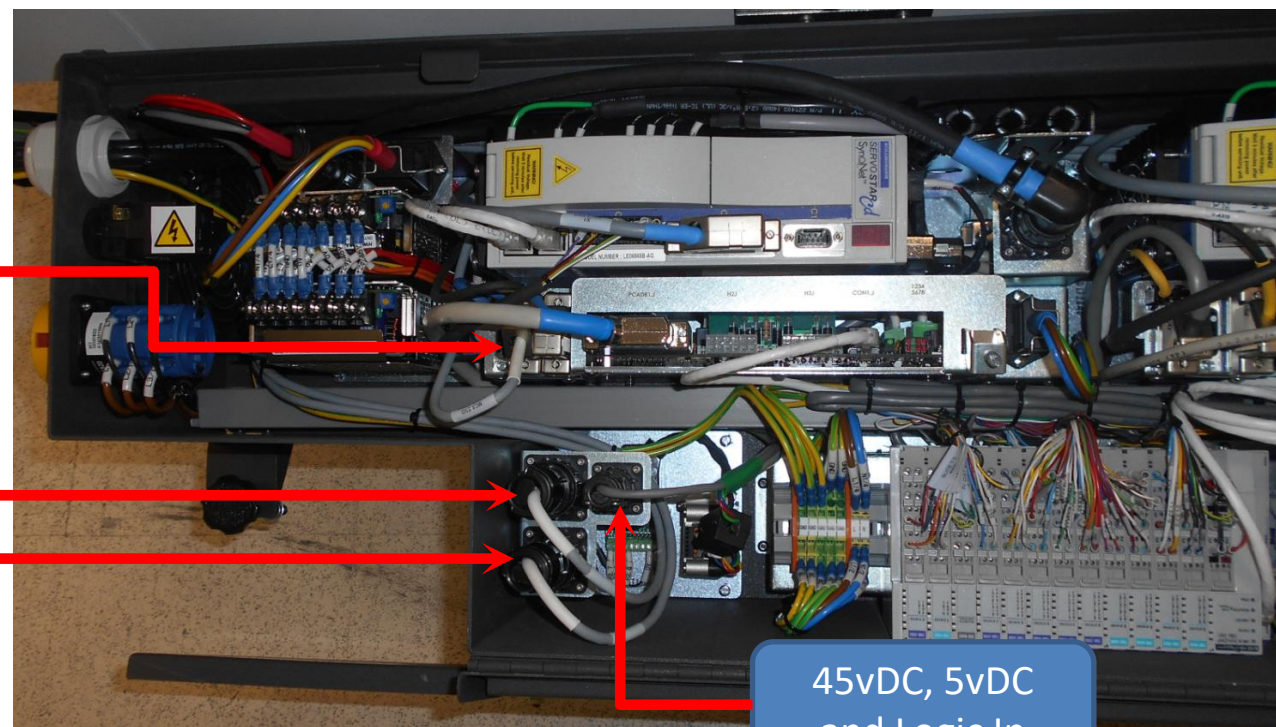
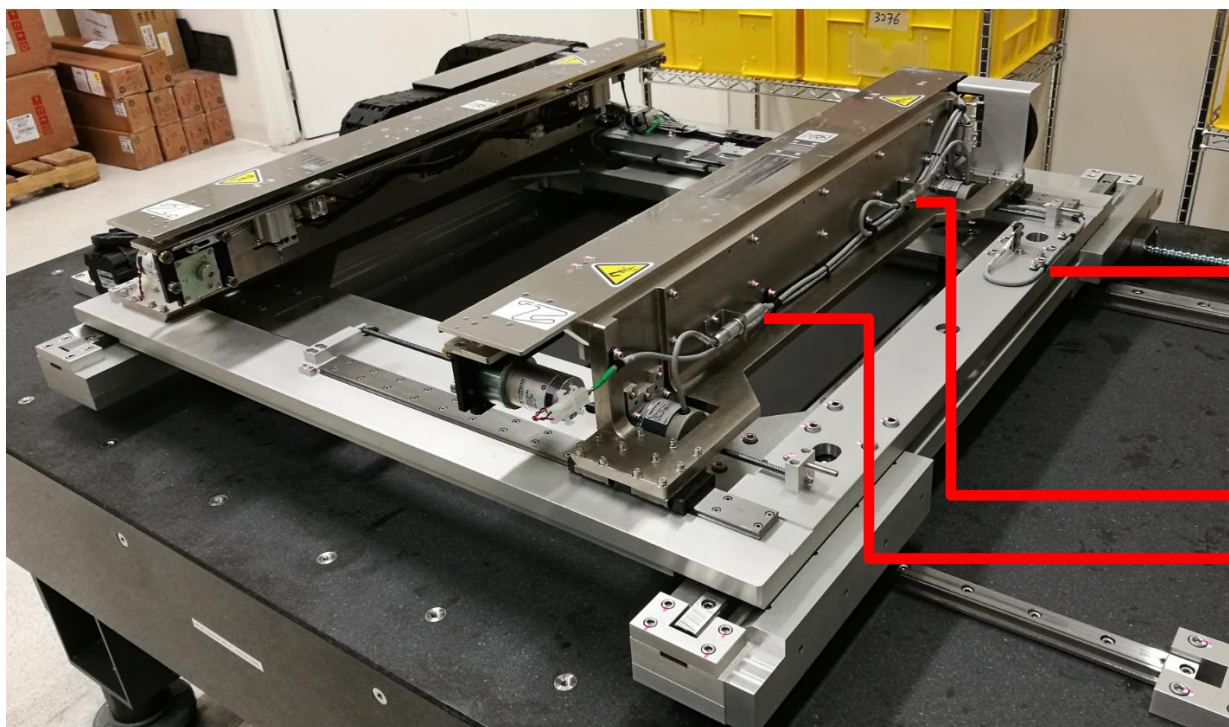
Fix Rail Conveyor Belt Routing



Adjustable Rail Conveyor Belt Routing

ABI Hardware Overview - Auto Width Adjustable (AWA)

Automated Board Inspection



45vDC, 5vDC
and Logic In

- Left and Right motor cable have no orientation as X-Koll assemble control rail width in parallel.
- 48vDC Lamba Power Supply (Configure as 45vDC) direct supply current to Stepper driver module
- 5vDC and Logic In supply from DIO Block

ABI Hardware Overview - Auto Width Adjustable (AWA)

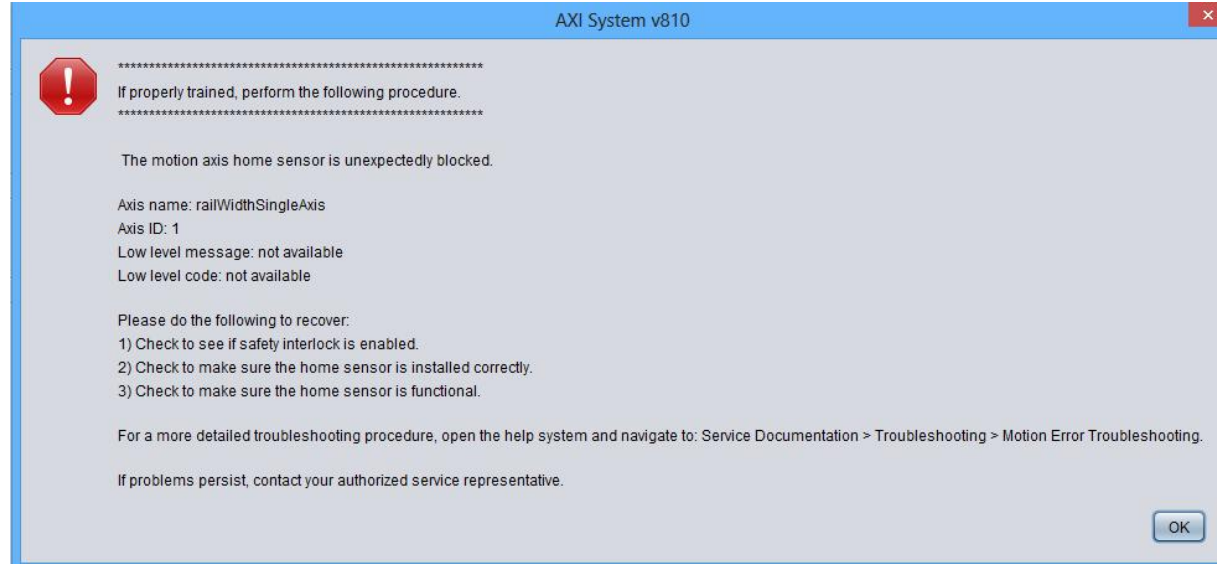
Automated Board Inspection



- Moves the Adjustable Rail to the correct distance from the Fixed Rail for the panel to be loaded into the system and inspected
- AWA have 2 stepper motor, each one at the Adjustable Rail end to drive the opening of the Adjustable Rail to correct width
- Have 2 lead screws
- 48 VDC supply is set to the both stepper motor

ABI Troubleshooting - AWA Homing Issue

Automated Board Inspection

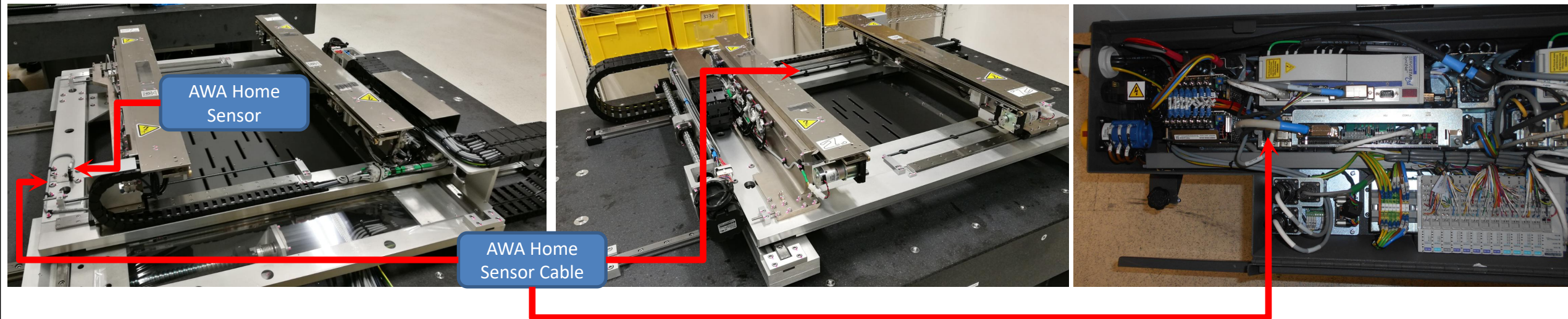


Potential Root Cause:

1. AWA home sensor misalignment/Mul-Function
2. AWA home sensor cable or Left or right AWA motor cable connectivity problem
3. X-Axis Kollmorgen Drive Kit, Servo interface card spoiled.

ABI Troubleshooting - AWA Homing Issue

Automated Board Inspection

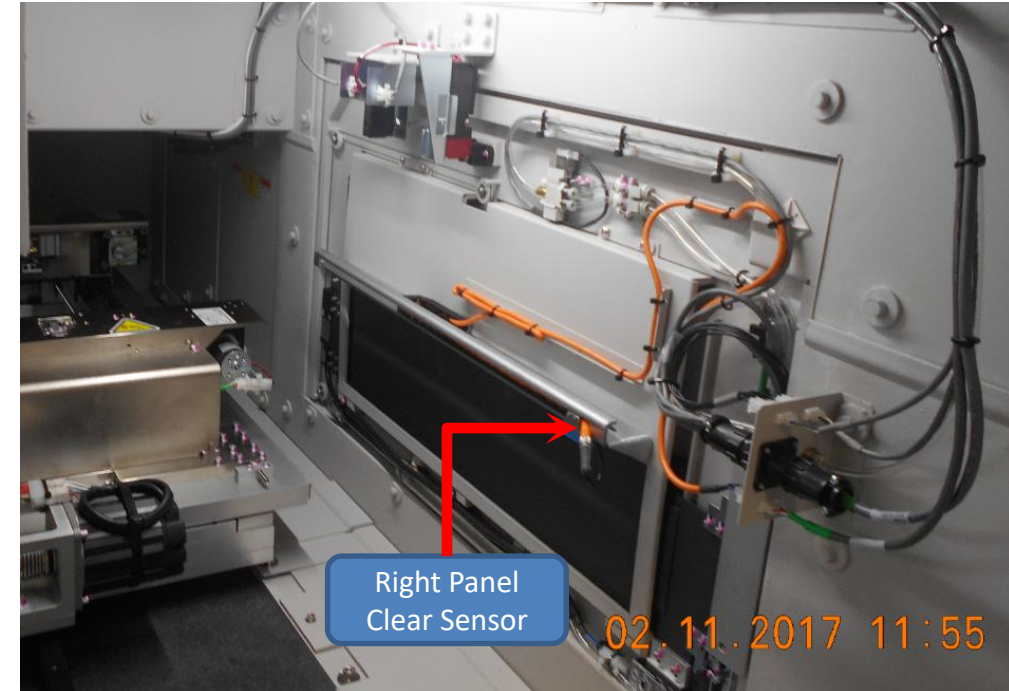
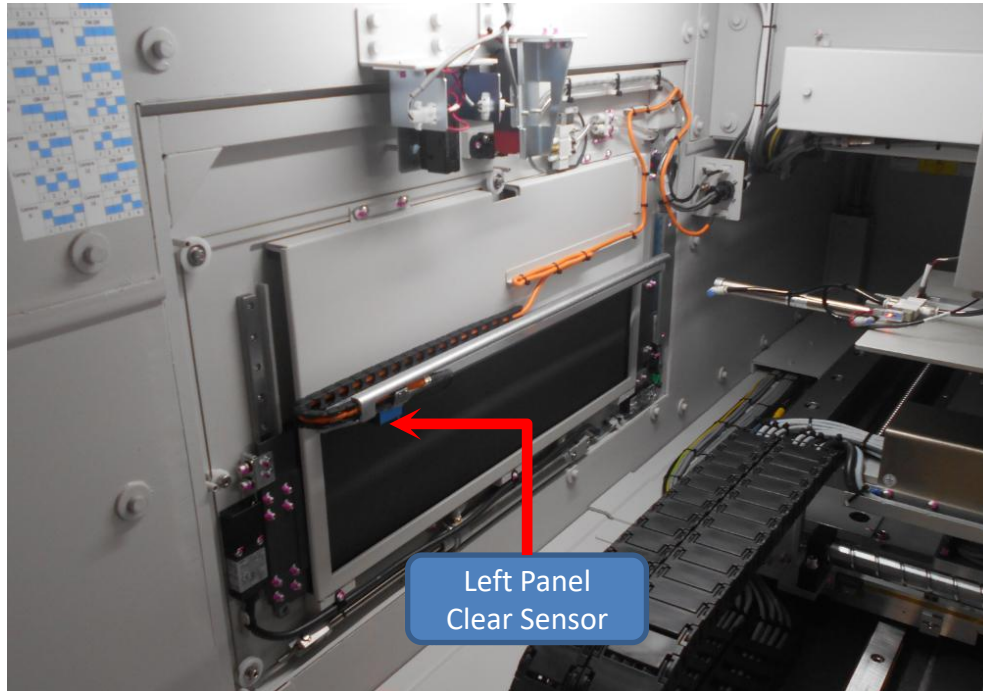


Troubleshooting Tips:

1. Remove and physical check on the home sensor and make sure no micro crack on sensor or wrong P/N used.
2. Check the cables connectivity. May refer to remove and replace procedure for technical drawing
3. Visual check on cables to ensure no snapping issued
4. Reseat the connectivity for AWA home cable, both end.

ABI Hardware Overview - Panel Clear Sensor

Automated Board Inspection

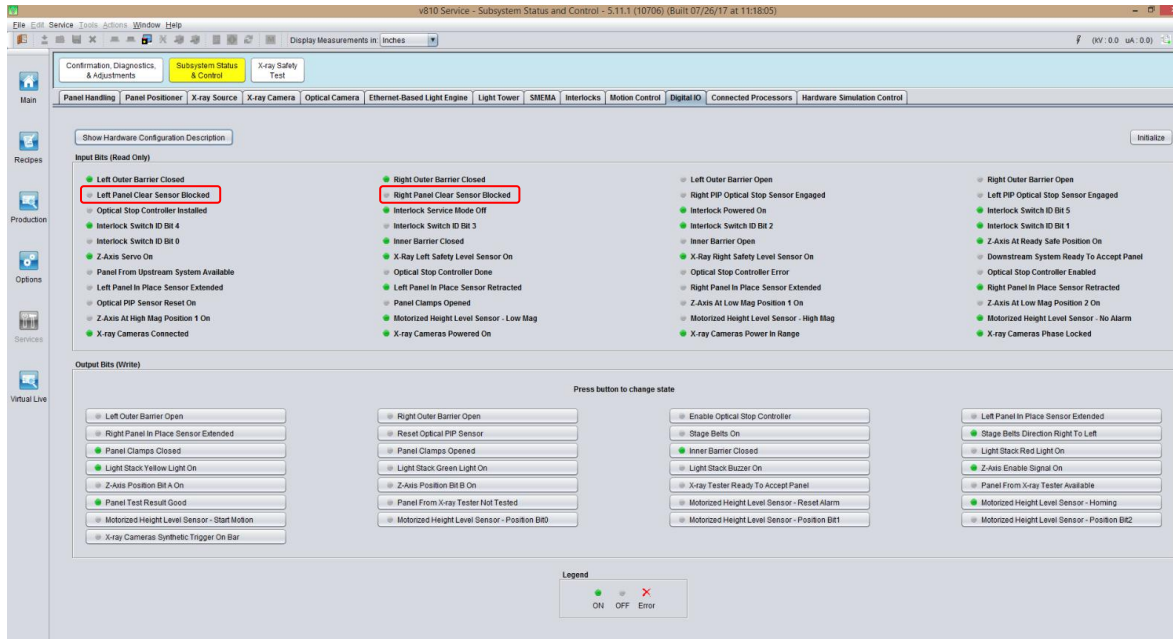


- To detect if any object present under sensor before XY table move
- To trigger conveyor belt on for board indexing
- S2ex machine type, Panel Clear Sensor did not have reflector
- Both left and right Panel Clear Sensor is mirror position
- The intensity/sensitivity of the Panel Clear Sensor is adjustable

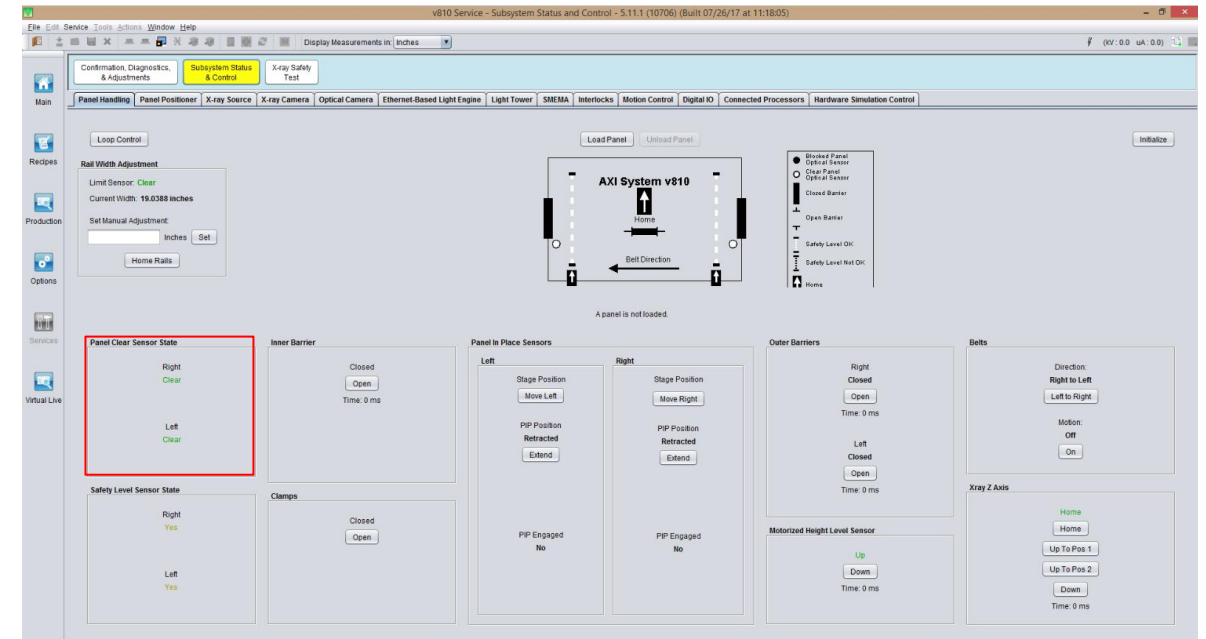


Hardware Overview - Panel Clear Sensor

Automated Board Inspection



DIO tab



Panel Handling tab

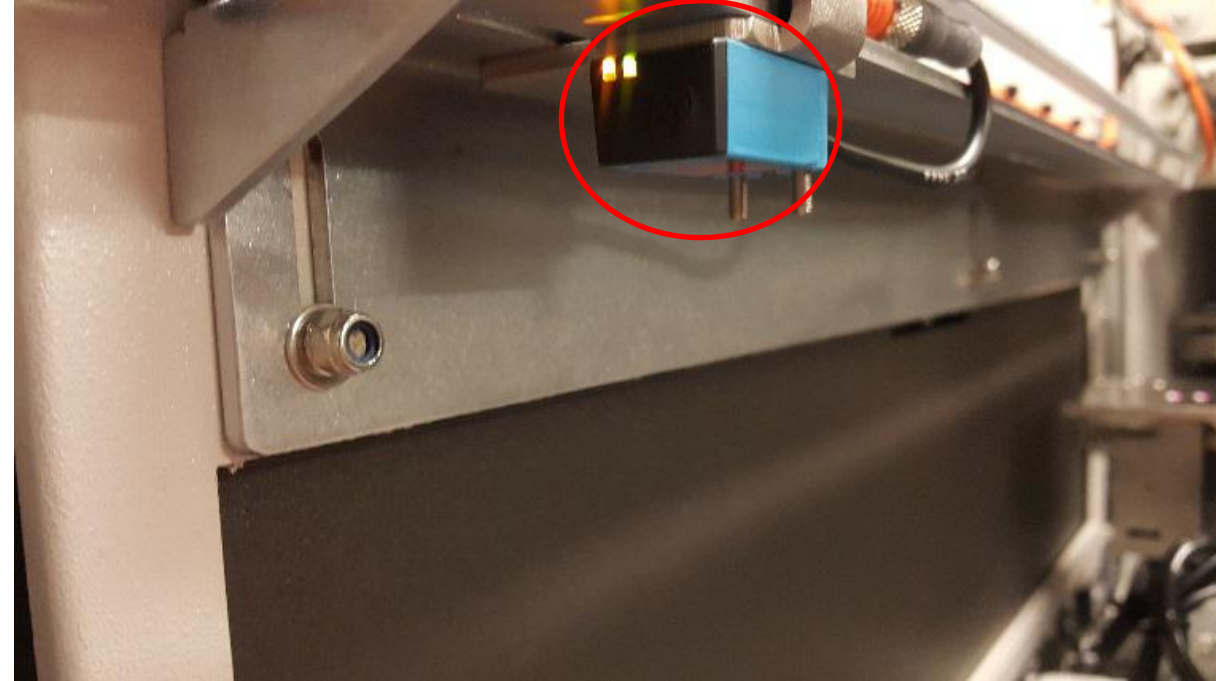
- Panel Clear Sensor should always in “Clear” status except board present on top conveyor belt surface (clamp or open).
- On Panel Handling tab, if Panel Clear sensor in “Blocked status” (Green Light), XY table unable to Home Stage.

ABI Hardware Overview - Panel Clear Sensor

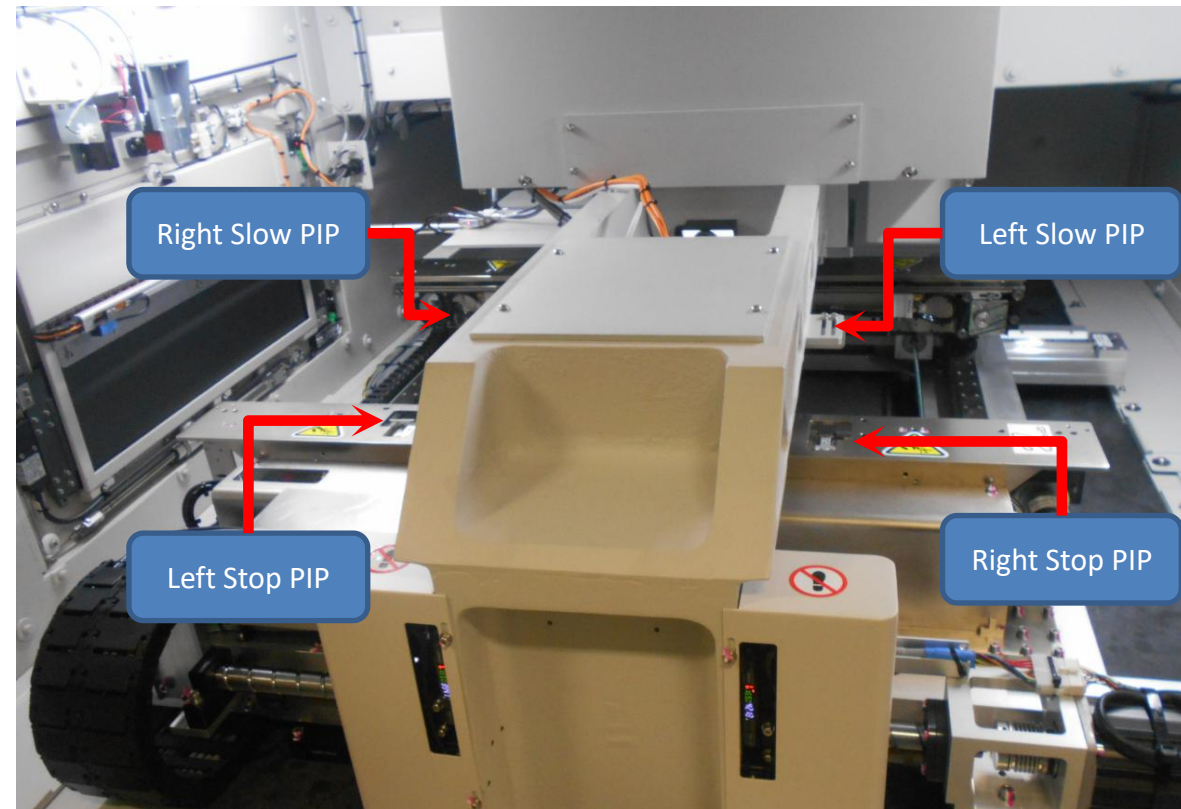
Automated Board Inspection



Panel Clear Sensor Clear Status



Panel Clear Sensor Blocked Status

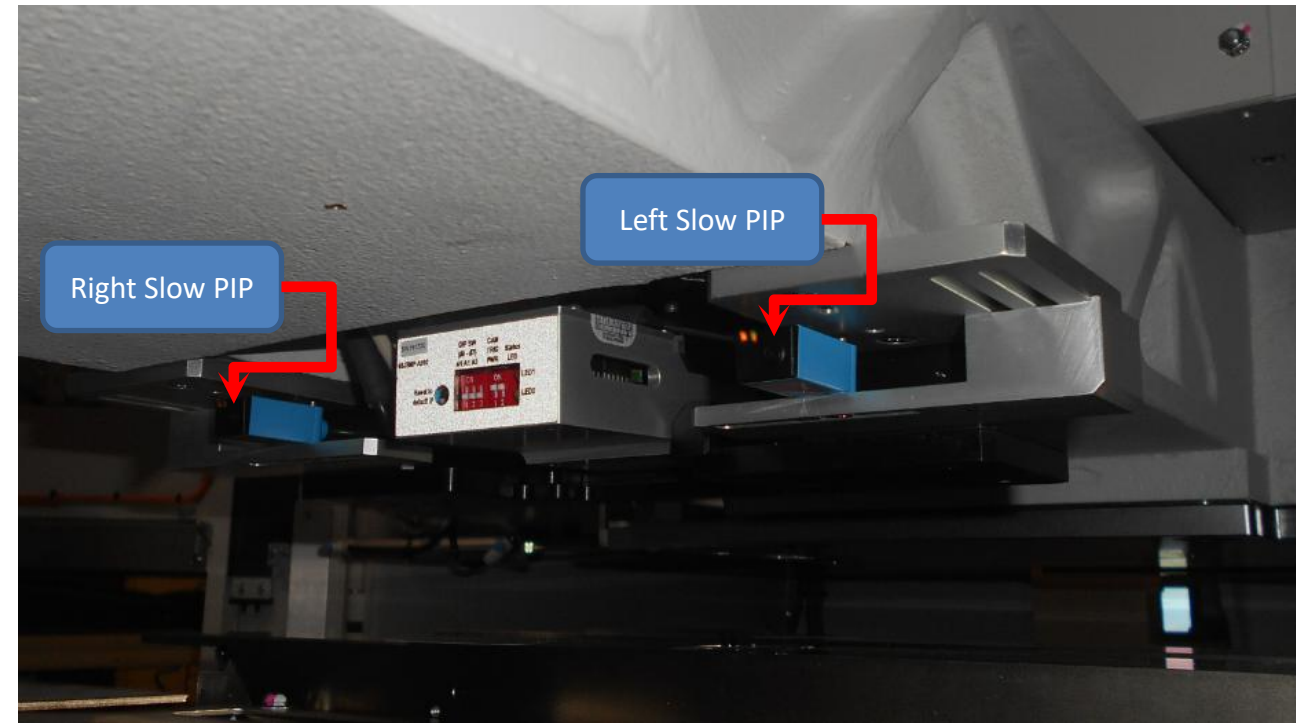
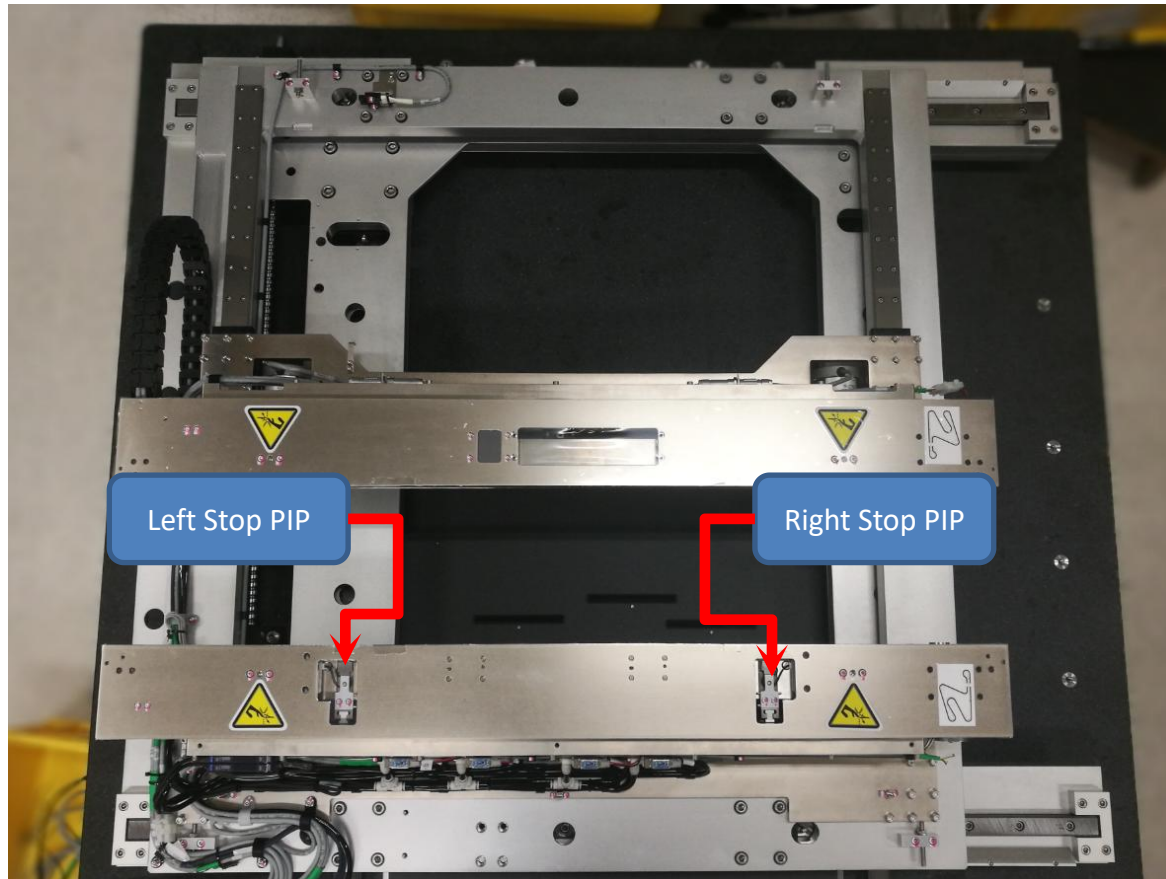


What is Panel In Place (PIP) Module?

- PIP serve as the module to stop the board at the accurate stopping position while loading the board
- Hybrid PIP (Optical Slow + Mechanical Stop) is used to replace the Rail Optical PIP to serve the same purpose.
- To trigger board stop after board indexing complete in position

ABI Hardware Overview - XY Stage - Panel In Place Sensor

Automated Board Inspection



ABI Hardware Overview - Hybrid Panel In Place Sensor

Automated Board Inspection

Teflon PIP Guide

Cover PIP Wire

PIP Cylinder Nose

Board Stop Mount

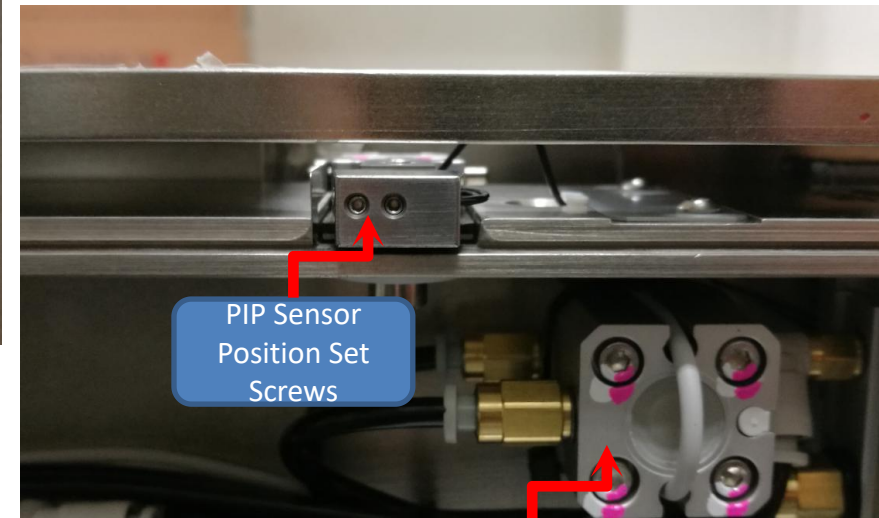
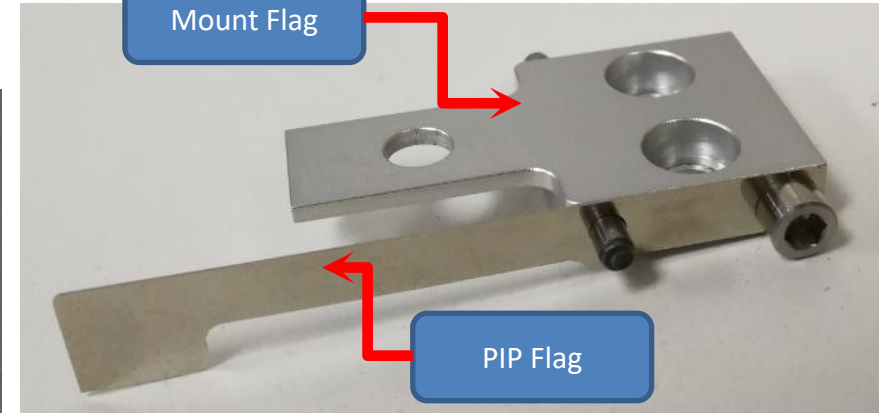
Fiber Optic PIP Sensor

Mount Flag

PIP Flag

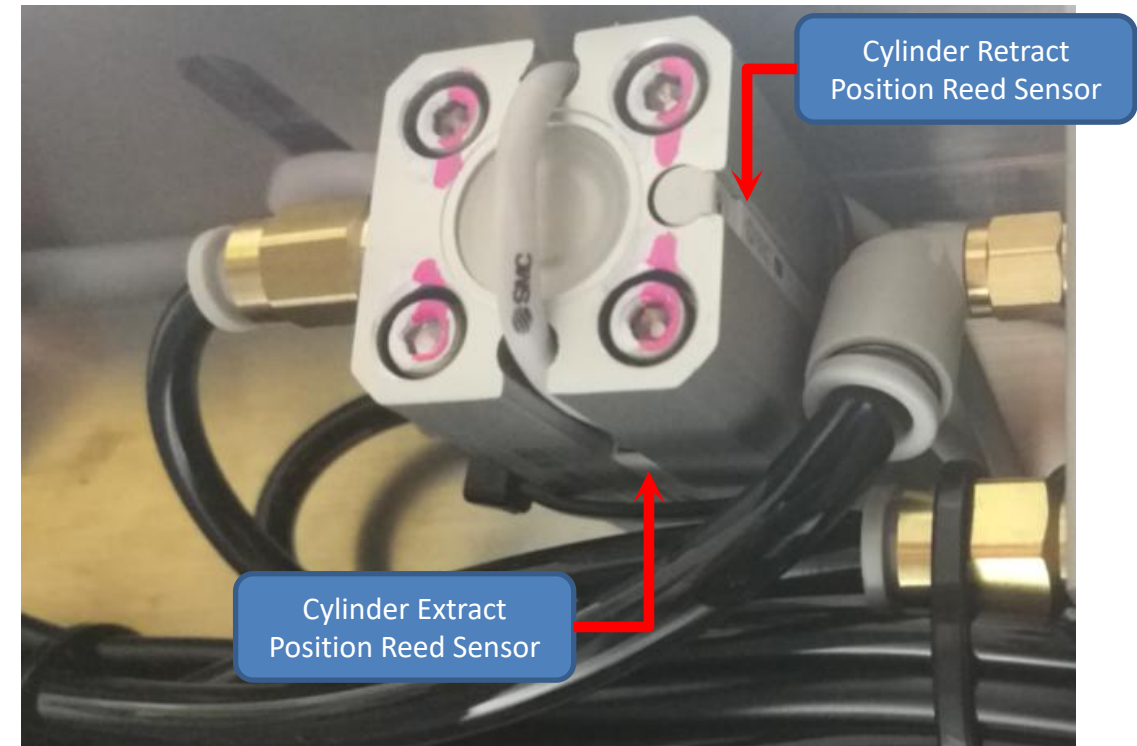
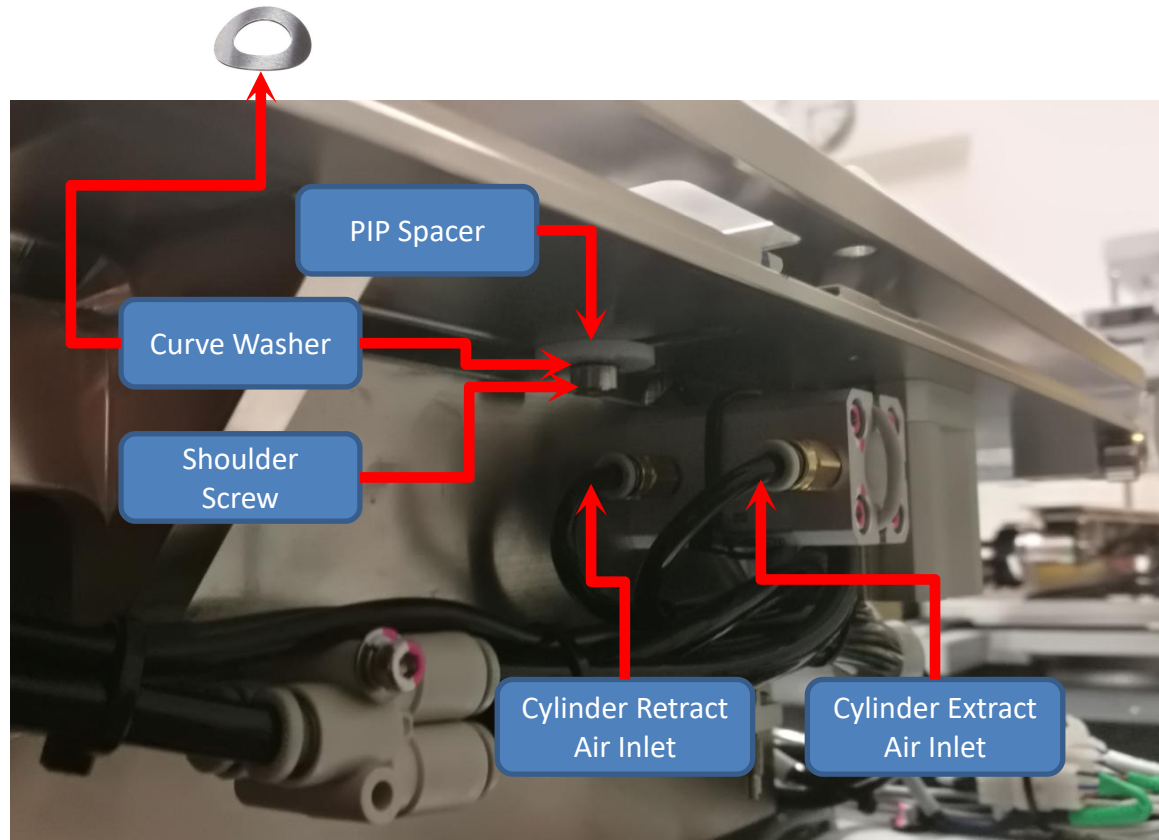
PIP Sensor Position Set Screws

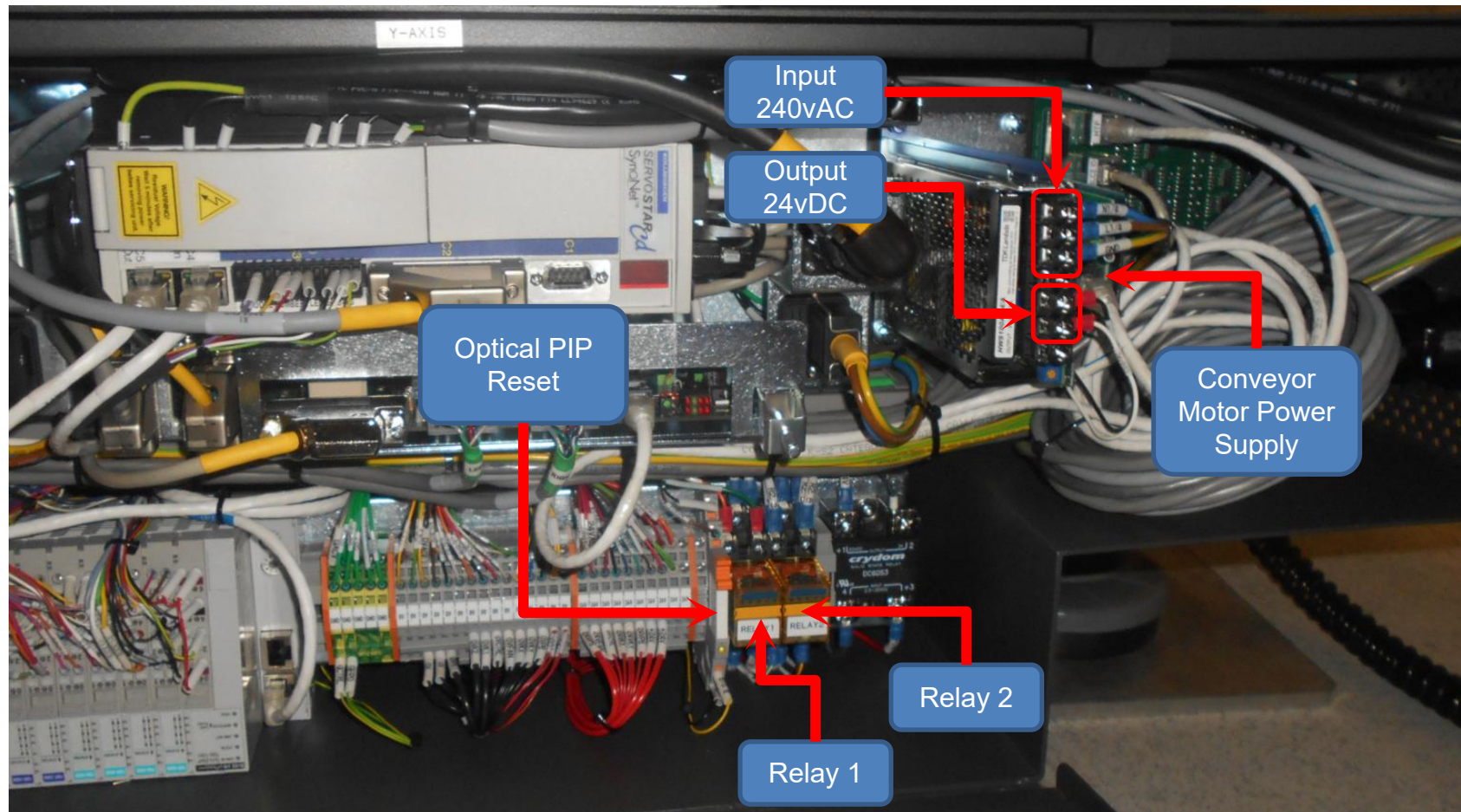
PIP Cylinder



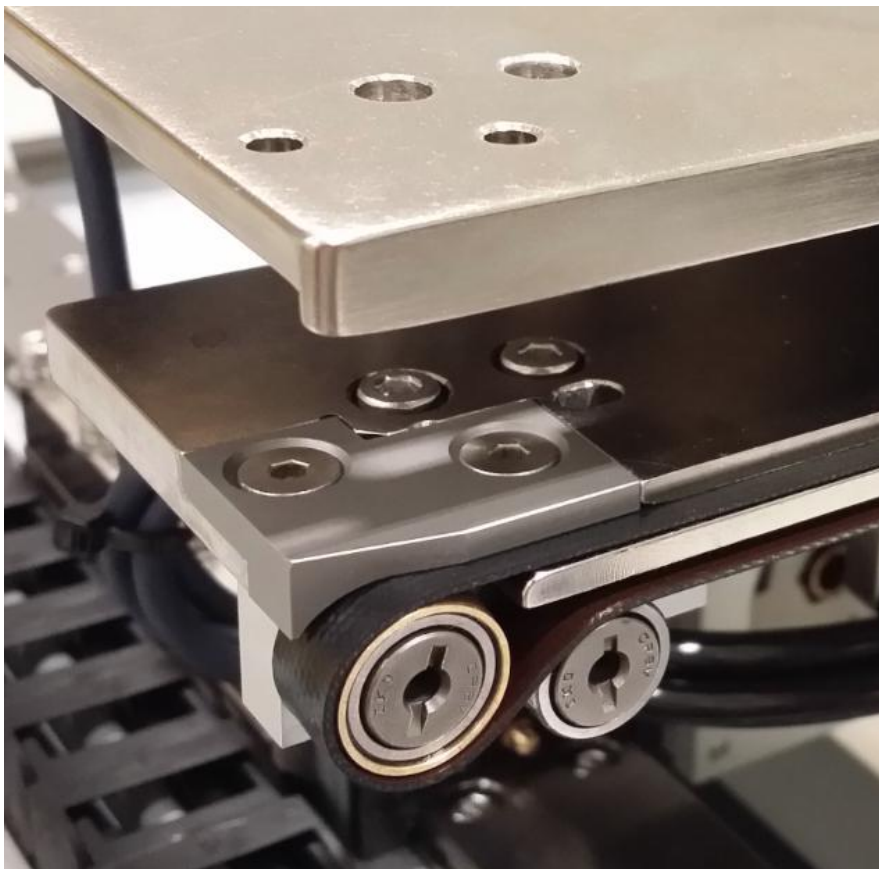
ABI Hardware Overview - Hybrid Panel In Place Sensor

Automated Board Inspection

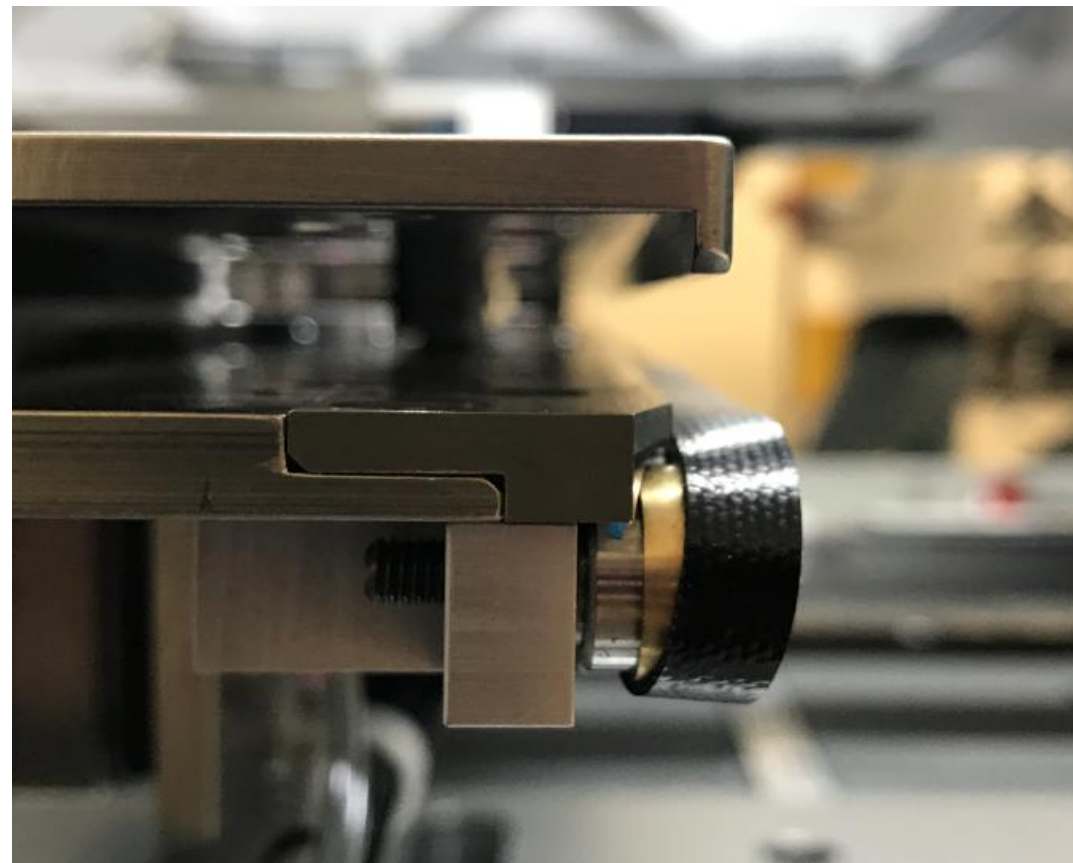




- Relay 1 = On/Off Conveyor Motor
- Relay 2 = Conveyor belt direction change (forward/revert)
- Optical PIP Reset Relay = Reset PIP Amplifier (Only applicable to Rail Optical PIP)
- Conveyor Motor Power Supply (24vDC) = supply current to both conveyor motor



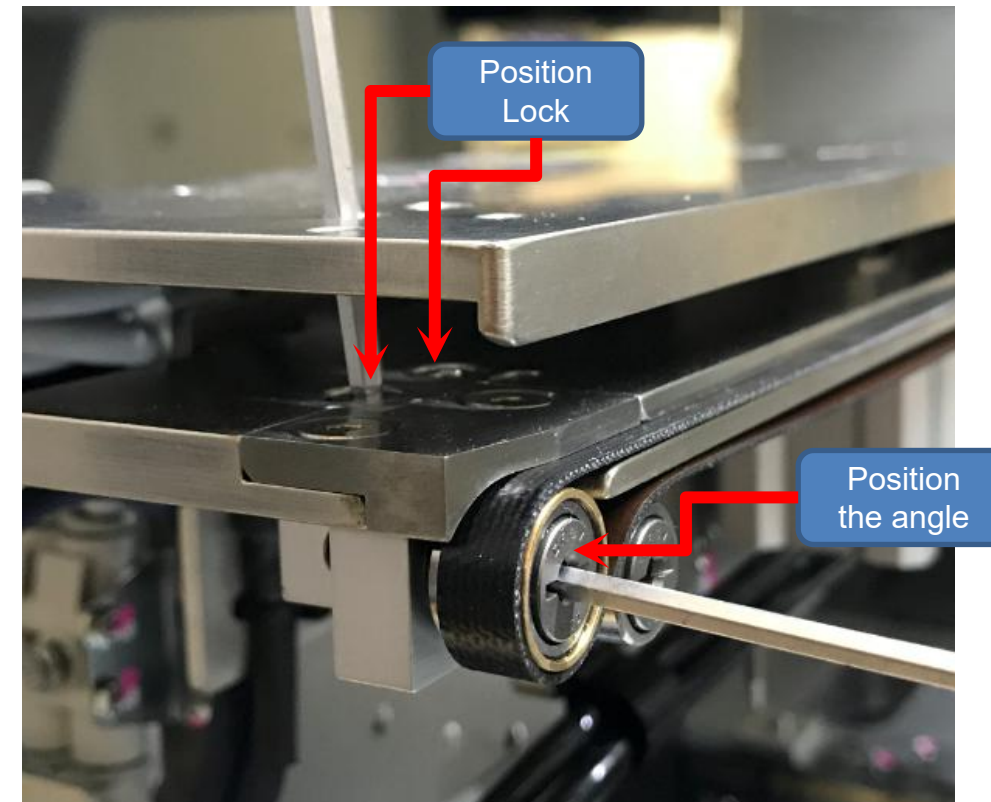
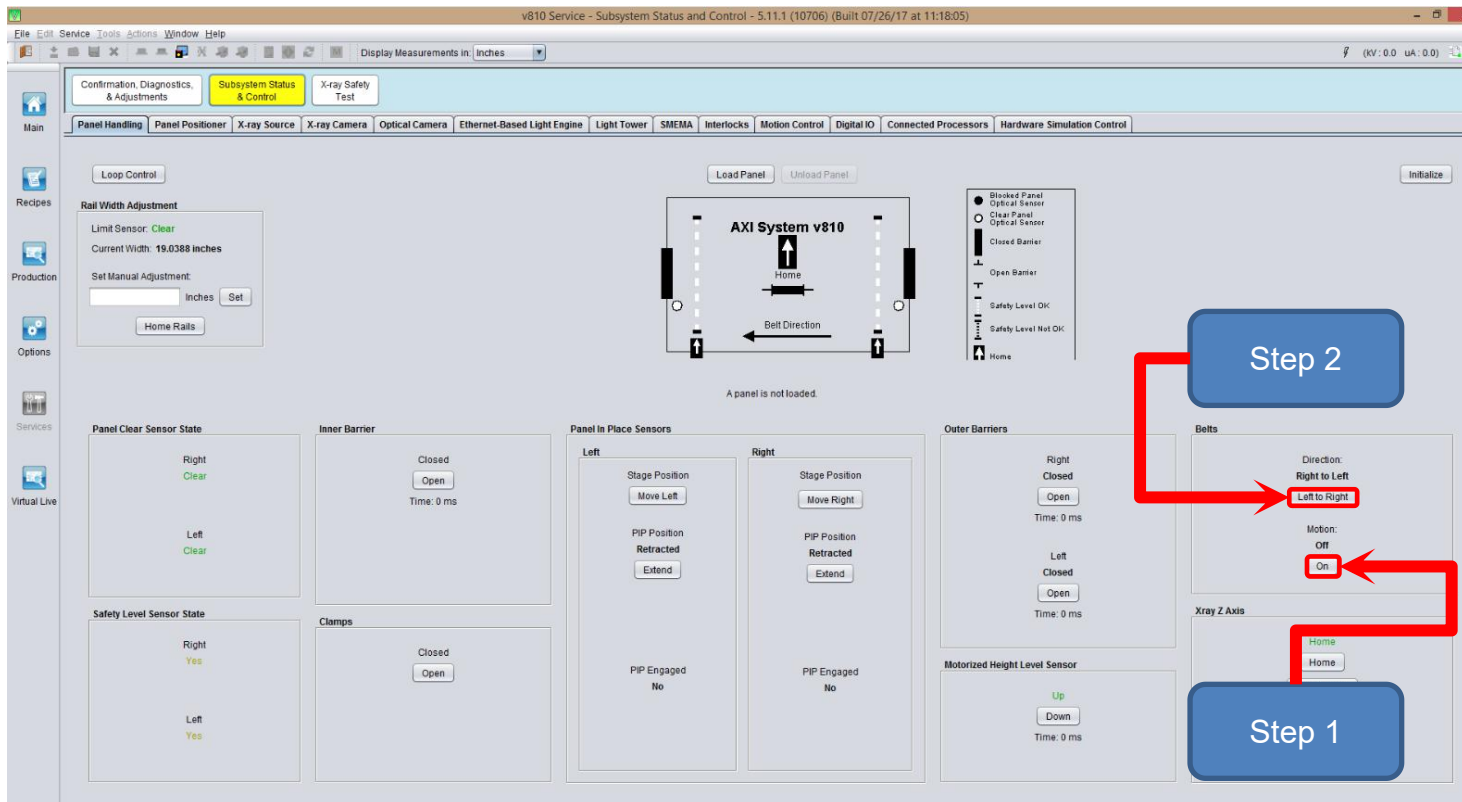
Belt in alignment



Belt not in alignment

ABI Hardware Overview - Conveyor Belt Alignment

Automated Board Inspection



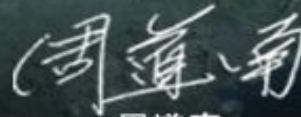
1. Turn the conveyor belt on
2. Reposition the End Bearing Block until the belt is spinning middle of the pulley kit and tighten the screws for position lock.
3. Run belt in forward and revert direction and validate the belt alignment



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